

Chapter 29. Mechanical Ventilation in the Acute Respiratory Distress Syndrome

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Mechanical Ventilation in the Acute Respiratory Distress Syndrome: Introduction

Few areas of critical care medicine have been the subject of as much investigative attention or clinical concern as the set of problems grouped under the label *acute respiratory distress syndrome* (ARDS). This syndrome, first formally described in 1967,¹ continues to be recognized clinically as a rapidly developing impairment of pulmonary oxygen exchange accompanied by diffuse infiltrates and altered respiratory system mechanics that cannot be attributed solely to hydrostatic forces. Fueled by better characterization of innate pathophysiology and of iatrogenic factors, considerable progress has been made in recent years toward reducing the adverse consequences of this condition. Yet, after more than four decades, active debate continues regarding key elements of the ventilatory prescription and appropriate therapeutic targets.

From the outset, mechanical ventilation with positive pressure has been essential in addressing the life-threatening gas exchange abnormalities and otherwise unsustainable workloads associated with ALI. Only in the relatively recent past, however, has there been clear documentation that the tidal pressures of mechanical ventilation can impact morbidity and survival.^{2,3} This awareness has caused a conceptual shift away from attempting to restore normal blood gases at the costs of high pressure and toward adopting the avoidance of preventable iatrogenicity (“lung protection”) as the first priority.

Many aspects of the debate concerning appropriate ventilator management of this group of conditions can be traced to the heterogeneity of the patient population, to our still imperfect comprehension of the mechanisms of ventilator-associated lung injury (VILI) and to the relative imprecision of the criteria upon which the label *ARDS and/or acute lung injury (ALI)* is assigned.⁴ Despite an incomplete and still evolving understanding, a rich experimental and clinical database—much of it collected over the past two decades—allows for the development of a rational set of principles upon which to formulate an effective ventilation strategy.^{5–7} Definitive answers for many important clinical questions related to this topic are not available; what is presented here reflects a pathophysiology-guided approach to accomplish essential clinical objectives while avoiding VILI (Table 29-1).

Table 29-1: Conceptual Principles in Acute Respiratory Distress Syndrome Ventilation

ARDS is a heterogeneous problem
Between patients
Over time
Between lung regions
Risk for ventilator-induced lung injury is proportional to <i>transalveolar</i> pressure
Lung recruitment is essential to avoid ventilator-induced lung injury
The chest wall influences regional lung volumes, tolerated pressures, and recruitability

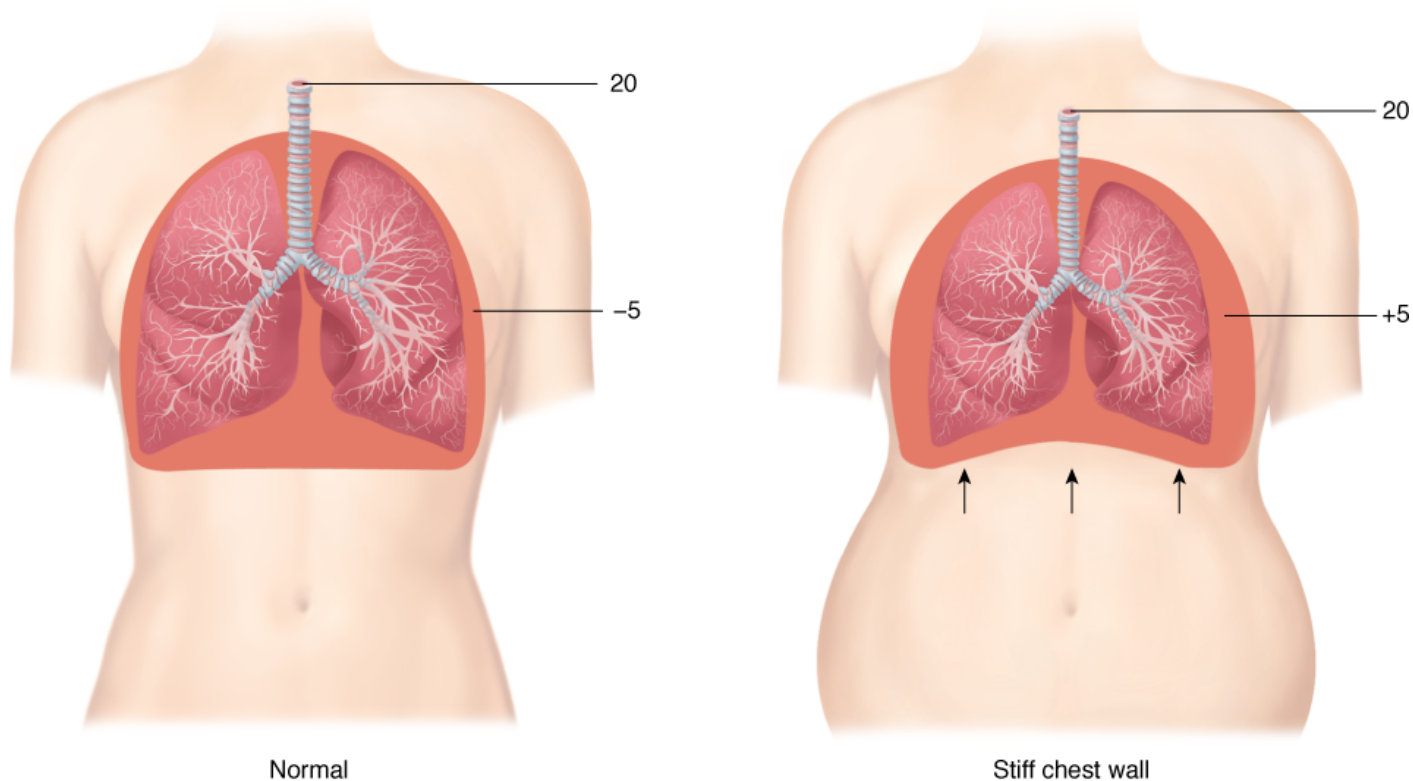
General Objectives for Ventilator Support

In the clinical setting, mechanical ventilation ensures adequate oxygenation of arterial blood, provides sufficient oxygen transport to vital organs and tissues, assists in eliminating carbon dioxide, relieves excessive burdens placed upon the respiratory muscles, helps maintain alveolar stability, and allows therapeutic measures that require controlled ventilation. Despite its undeniable value, however, mechanical ventilation also has the potential to inflict adverse clinical outcomes. The task of accomplishing ventilation safely in patients with injured lungs is made far more difficult by the mechanical heterogeneity of the respiratory system and the diversity of pathophysiology encountered among different patients who satisfy extant operational criteria for this condition.

The terms *ALI* and *ARDS* comprise a category of patients with varied pathoanatomy and mechanical characteristics. According to the widely used American-European consensus guideline,⁴ the primary criteria relate to pulmonary oxygen exchange, the appearance of the plain chest radiograph, and to a clinical assessment of left ventricular function. Although such broadly inclusive criteria may be useful for some purposes, they prove problematic for others. In formal definitions, for example, no provision is made for the level of positive end-expiratory pressure (PEEP) at which the arterial sample is obtained or the chest radiograph is exposed. No stipulation requires that the defining criteria be met under standardized conditions and remain reproducible over time. Chest wall properties and body weight are left unaccounted for. Yet, pulmonary oxygen exchange is influenced not only by the properties of the lung, but also by end-expiratory lung volume, body position, mixed venous oxygen content, pulmonary blood flow, and the integrity and intensity of hypoxic pulmonary constriction. The chest radiograph is interpreted subjectively, and expert assessments often differ.⁸ Moreover, it is clear that the lungs of different patients with ARDS vary with regard to radiographic appearance, inherent recruitability, and histopathology. Even within the same individual, assessed at the same moment, the pathoanatomy and mechanical environment varies from site to site within the injured lung. Such regional differences are partially explained by the properties of the surrounding chest wall, which profoundly influence the inflation characteristics of the integrated respiratory system as well as their regional distribution.^{9,10}

Similar clinical presentations can mask radical differences in lung pathology, mechanical properties, and response to ventilator settings and maneuvers. Lungs of patients with ARDS resulting from pneumonic consolidation, for example, are less likely to inflate easily than are lungs made edematous by the circulating mediators of extrapulmonary sepsis.¹¹ Computed tomographic (CT) patterns may reflect these differences.¹² Moreover, inflexibility of the chest wall may substantially increase the pressures required to inflate the respiratory system (Fig. 29-1).¹³ From these considerations, it is clear that inflexible guidelines for selecting PEEP and tidal volume that advise specific numerical values of these settings will be variably effective in accomplishing the goal of minimizing tissue stresses, depending on the type and severity of lung injury, the compliance of the chest wall, and the ranges for opening pressures and closing tendencies among the multiplicity of the lung units that comprise the injured lung.

Figure 29-1



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Influence of chest wall compliance on transpulmonary pressure and lung volume. Any specified airway pressure is associated with less transpulmonary pressure in the presence of chest wall stiffness or expiratory effort.

Pathophysiologic Features Relevant to Ventilatory Support

A major conceptual advance of the past 25 years is the recognition that the lungs of patients with ARDS are mechanically heterogeneous and vary enormously in their patterns of infiltration, their inflation and collapse properties, and their regional expressions of pathology. CT scanning has

Loading [Contrib]/a11y/accessibility-menu.js or diffuse—with implications for response to ventilator interventions such as PEEP.¹² Variation in the

underlying conditions that give rise to the clinical problem described as ARDS precludes categorical histologic descriptions that apply across the full range of clinical experience. A few salient characteristics, however, are shared by most. In its earliest phase, noncardiogenic (high permeability) edema gives rise to a lung whose parenchymal airspace is partially occupied by proteinaceous edema and cellular infiltrate. A relatively large proportion of the lung—often exceeding 50%—is airless at end expiration, with the exact percentage depending jointly on severity, disease stage, etiology, and PEEP. Destruction of surfactant-producing type 2 alveolar cells leads to its diminished production, while exuded proteins and inflammatory products compromise the viability of the surfactant that remains.¹⁴ Surfactant plays several important biologic and physiologic roles. From a purely mechanical standpoint, the loss of functional surfactant increases surface tension, thereby contributing to alveolar flooding, increased tissue elastance, small airway closure and atelectasis—particularly at low lung volumes.

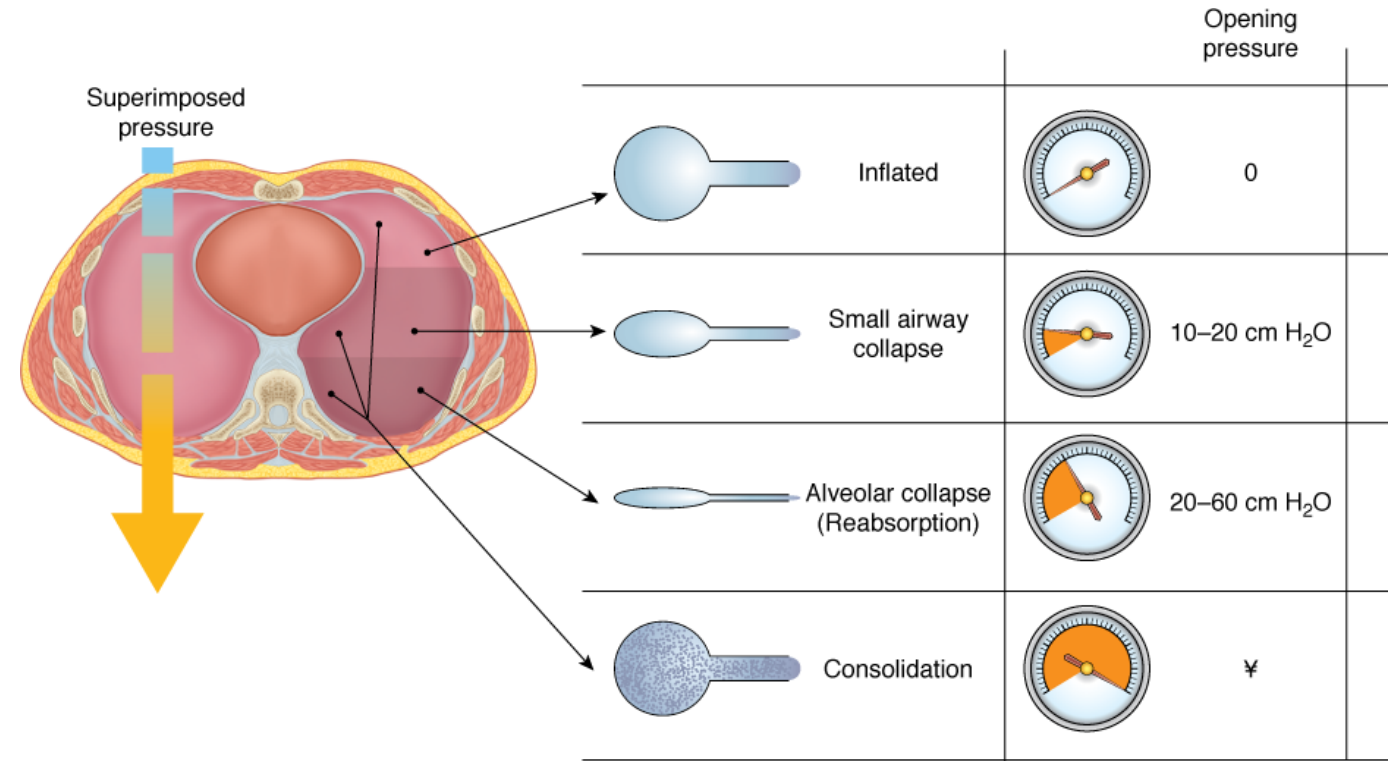
The relative proportions of airless and aerated tissue also vary with disease type and stage. Although some controversy persists, CT scan estimates of gas and tissue volumes suggest that potentially “recruitable” tissue comprises only a minority of radiographic density at functional residual capacity in most cases.¹⁵ Flooded and consolidated lung units comprise the remainder. Although clearly a minority viewpoint, a plausible argument has been advanced that “reopening” of lung units by alveolar pressure may occur primarily by redistributing alveolar liquid and shifting fluid volume from the alveolar to the interstitial compartments of the lung—not by atelectasis reversal (“recruitment”).¹⁶ The rapidity with which CT tissue density develops and resolves, however, as changes of alveolar pressure are imposed, as well as direct observations by intravital microscopy of alveoli at the lung’s surface,¹⁷ cast doubt on the primacy of the “fluid-shift hypothesis.”

Mechanical Properties of the Injured Lung

Replacement of airspaces by inflammatory debris, cells, and fluid results in a lung whose aeratable capacity and compliance (measured in terms of volume accepted per unit of pressure) is severely reduced. The compliance of the respiratory system (lungs and chest wall) falls during ARDS for two reasons. First, airspace is lost to fluid and cellular infiltrates. Second, many functional lung units operate near their elastic limit because many fewer are available to accept the tidal volume. Under normal conditions, surfactant-modified surface forces allow the lung to inflate and deflate at similar pressures. In contrast, the injured lung is characterized by a right-shifted pressure volume loop, made so by its reduced aeratable capacity and by surfactant depletion.^{13,18,19} For the same tidal volume, therefore, the mechanical work of breathing is dramatically increased.

Although clinicians characterize the mechanics of the injured lung by airway pressure and flow measurements made at the airway opening—the common entry and exit point for gas exchange with the environment—the mechanical properties of the lung’s individual subunits are hardly uniform, even in health. In part, this heterogeneity relates to regional variations of pleural (and therefore transpulmonary) pressure that arise from interactions of the chest wall with the injured lung and from the need for dependent tissues to support the weight of the mediastinal contents and the edematous lung (Fig. 29-2). This underlying mechanical heterogeneity is implied by quantitative CT imaging techniques that characterize the topographical anatomy in response to changing patterns of airway pressure, or more directly in real time by imaging of ventilation with radiotracer gases or electrical impedance tomography.^{20,21}

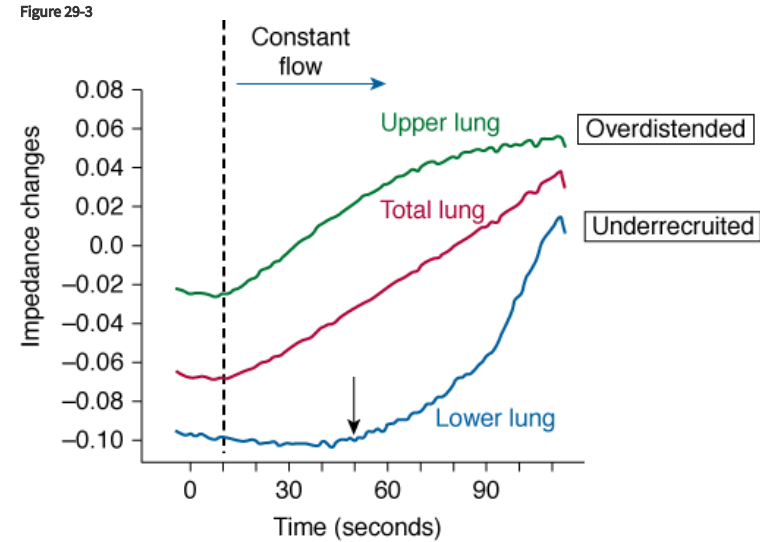
Figure 29-2



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Spectrum of opening pressures associated with lung units within the injured lung. Lung units located in dependent areas are compressed by the weight of the overlying lung and mediastinum. Local transpulmonary pressures vary considerably, so that some lung units remain fully inflated throughout the tidal cycle, whereas others cannot be aerated. Less pressure is required to open small airways when the alveoli remain aerated than when all alveolar gas has been absorbed. (Image provided by Luciano Gattinoni and used with permission.)

For any specified airway pressure, some lung units are closed and others are open. Among the population of open units, there exists a range of states of lung unit expansion, depending on the local transpulmonary pressures that distend them. Even at airway pressures that are generally considered modest, some of the open lung units verge on overdistension, while others are on the compliant portion of their pressure volume relationship (Fig. 29-3). Alveolar distension that approaches the elastic limit stimulates surfactant production, but the repetitive application of nonphysiologic stretching forces, as during high tidal volume or high-pressure tidal ventilation, initiates molecular signaling of a local inflammatory response.^{22,23}

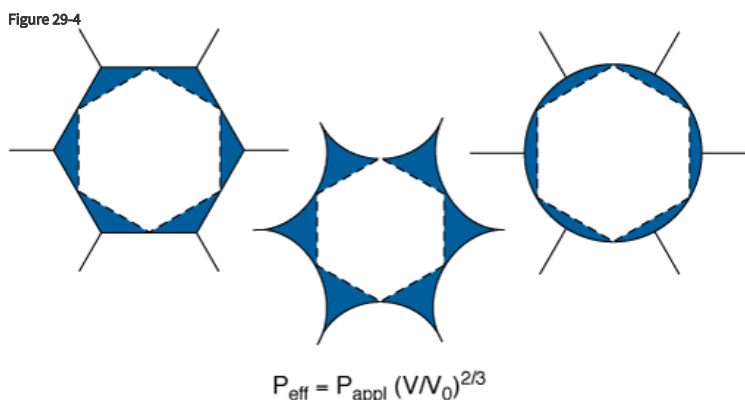


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Mechanical characteristics of lung units in dependent (*lower*) and nondependent (*upper*) lung regions. As the lung is inflated by constant flow of gas, electrical impedance increases in proportion to aerated volume. Although the relationship of volume to impedance for the total lung appears roughly

linear, upper lung regions are relatively overdistended whereas lower lung regions are relatively underrecruited until total lung capacity is achieved. (Image provided by Marcelo Amato and used with permission.)

Mechanical interdependence among the lung units of a heterogeneously affected lung amplifies the stresses of high pressure at the junctions of closed and open lung units in a nonlinear fashion (Fig. 29-4).^{23,24} Such recurring forces at the points of stress focusing not only strain the lung's structural meshwork to initiate inflammation, but also assist in reopening potentially recruitable (atelectatic) lung units.²⁵ When atelectatic tissues are subjected to high pressures, amplified shearing forces within these zones at high risk for damage may be of sufficient magnitude to tear the delicate terminal airways or alveoli themselves, creating micro wounds that produce tissue hemorrhage and incite inflammation as a secondary phenomenon. Stress amplification at the margins of dissimilar tissues is at least partially a function of their relative volumes. (A flooded, gasless alveolus of volume similar to its air-filled neighbors may experience forces that are no more intense.) Because mechanical forces are amplified at the junctions of dissimilar tissues, it seems likely that injury abets the process of ventilator associated injury, so that damage propagates outward from injured into previously healthy zones. Whether injury results from the process of repetitive popping open of lung units under high pressure or simply is the cumulative result of high tension stretching at the interface remains to be resolved. It appears, therefore, that there are at least two critical elements that place the injured lung at risk for ventilation associated lung damage: high inflating pressures and the prevalence of collapsed lung units that interface with open ones.



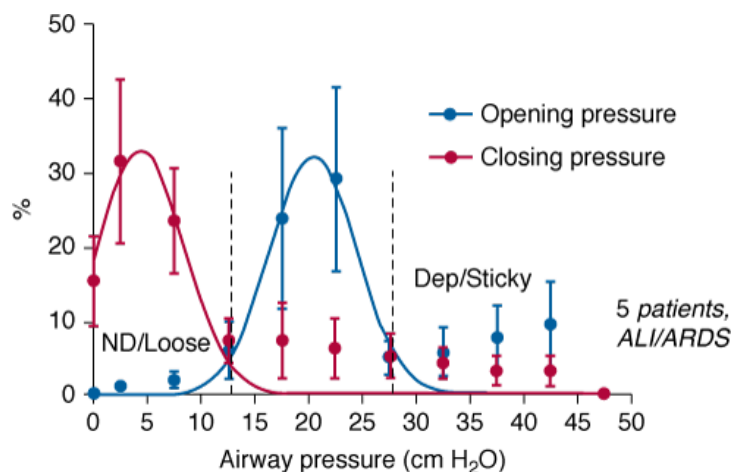
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Amplification of tension at the boundaries of open end collapsing lung tissue. Normally distended (*left*), collapsing (*center*), and overdistended (*right*) states of inflation. Because of interdependence, the tensions applied to a collapsing alveolus are amplified. A simple mathematical model predicts that the tensions resulting from an alveolar pressure (P_{appl}) produce a tension comparable to an effective pressure (P_{eff}) described by the equation. At 30 cm H₂O applied pressure, the volume of an aerated alveolus (V) is approximately ten times that of a collapsed alveolus (V_0). Thus, the estimated pressure necessary to mimic the tensions at the interface is approximately 4.5 times as great as that applied, or 140 cm H₂O. (Used, with permission, from Mead et al.²³)

Atelectasis

At the microscopic level, lung unit collapse may be thought of in two broad categories: “loose” atelectasis that arises primarily from compressive forces of the heavy lung acting to close small airways and responds to relatively low levels of transpulmonary pressure, and adhesive or “sticky” atelectasis that results from gas absorption and requires very elevated pressures to reverse.²⁶ High concentrations of inspired oxygen encourage the latter in poorly ventilated regions, as well as interfere with surfactant kinetics. These hard to recruit units may be among the first to close as high airway pressure is withdrawn. Loose and sticky types of atelectasis often coexist, perhaps accounting for the fact that lung units open throughout the entire range of total lung capacity (Fig. 29-5).^{25–27} In the setting of severe ARDS, pressures that exceed 60 cm H₂O may be needed to complete the recruiting process. Although opening of lung units is primarily a function of transalveolar pressure, the duration with which pressure is applied also contributes; that is, a lower pressure may be sufficient if applied for an adequate period. Moreover, atelectatic lung units tend to “yield” (open) in a stuttering, discontinuous fashion, with high sustained pressure causing a serial snapping open of small blocks of units.²⁸ Discontinuous lung opening during the tidal cycle may give rise to audible crackles, usually best heard in dependent regions.

Figure 29-5



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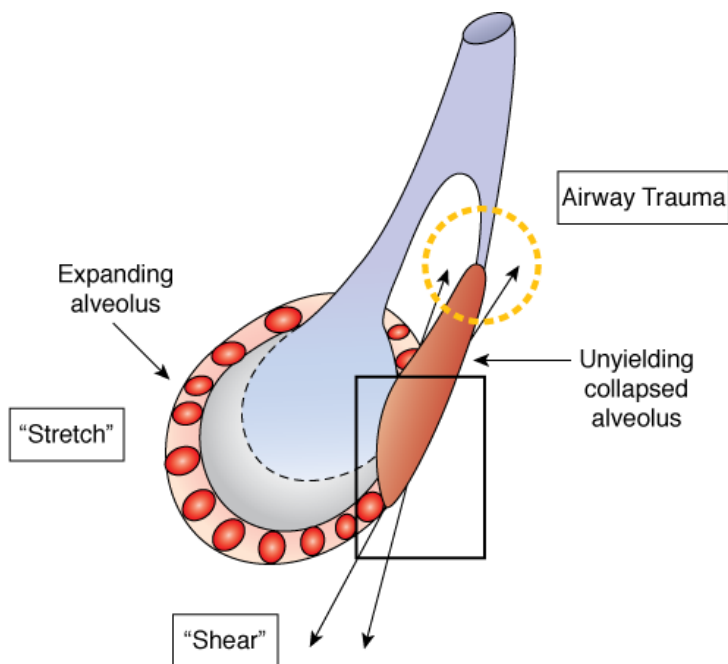
Histograms relating percentage of potentially recruitable lung units that are open to inflation and deflation airway pressures. Unstable airways open at relatively modest pressures, whereas alveoli that have undergone absorption collapse require much higher pressures to open. The spectrum of closing pressures is left shifted. Note that some lung units begin to collapse at relatively high airway pressures. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. Crotti S, Mascheroni D, Caironi P, Pelosi P, Ronzoni G, Mondino M, Marini JJ, Gattinoni L. Recruitment and derecruitment during acute respiratory failure. A clinical study. *Am J Resp Crit Care Med*. 2001;164(1):131-140. Official Journal of the American Thoracic Society. Adapted from original article.)

Pulmonary Edema

The increased microvascular permeability of ALI and/or ARDS renders it vulnerable to edema formation and impedes its relative rate of edema clearance. The weight of the normal lung does not begin to rise significantly until pulmonary venous pressure exceeds 20 mm Hg. Unlike the healthy lung, however, extravascular water in the injured lung bears a strong quasilinear dependence on hydrostatic microvascular pressure, a relationship that begins from capillary pressures that are considered low by most clinical standards.²⁹ The term *noncardiogenic pulmonary edema*, therefore, does not imply that gas exchange cannot improve by lowering the hydrostatic pressure gradient across the injured lung. Because extraalveolar microvessels as well as capillaries can leak,^{30,31} numerous factors influence the vascular pressures relevant to edema formation. These include left ventricular filling pressure, cardiac output (which increases the pressure needed to drive blood flow through the lung with limited capillary recruiting reserve), the plasma oncotic pressure, the interstitial fluid pressure, and the integrity of the alveolar and lymphatic lung water clearance mechanisms. Clearance mechanisms are influenced by the degree of lung stretch. Translocation of blood from infradiaphragmatic vascular capacitance beds to the central vessels of the thorax can occur as intraabdominal pressure rises.³²

Mechanisms and Consequences of Ventilator-Induced Lung Injury

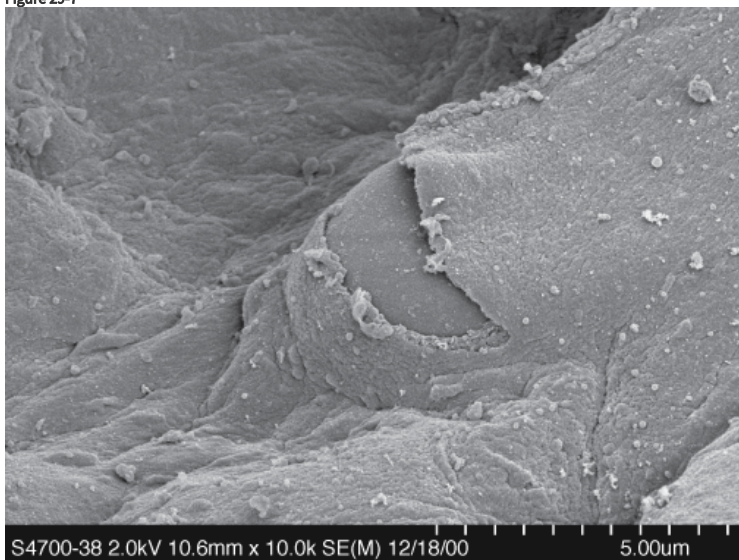
For more than three decades, experimental studies have shown that *excessive mechanical stresses* developed during mechanical ventilation can inflict injury upon both normal and acutely injured lungs, and once injured, retard their healing.^{33,34} Injury may result from overstretching of tissues that are already open, from shearing forces at the junctions of expandable and unyielding tissue, or from repeated percussion of closed terminal airways (Fig. 29-6). Repeated application of transalveolar pressures exceeding those corresponding to the inflation capacity of a healthy lung may disrupt the alveolar epithelial barrier, especially in the absence of sufficient end-expiratory pressure to reduce stress focusing by holding open mechanically unstable lung units.^{33,35} From a theoretical standpoint, sustained recruitment reduces the potential for damaging forces to concentrate at the boundaries of inflating lung and unyielding or collapsed structures. Indeed, an overwhelming body of experimental work indicates the lung protective effect of sustained “recruitment” when high tidal inflation pressures are used.^{36,37} Mechanosignaling at moderately high airway pressures may induce the formation and release of inflammatory mediators,^{22,37-42} initiate programmed cell death (apoptosis), or produce necrosis.^{38,41} Excessive strain may exceed cytoskeletal tolerances, causing physical tears and stress fractures of endothelium, epithelium, and intercellular matrix (Fig. 29-7).^{38,39} Scientific data suggest that when the ratio between tidal volume and aerated functional residual capacity exceeds 2, the resulting transalveolar forces are sufficient to cause tissue damage.⁴⁰ Furthermore, the tendency for translocation of inflammatory mediators and bacteria into the bloodstream is influenced by the ventilator pattern.⁴²⁻⁴⁴ Such observations suggest possible links between adverse patterns of ventilation and dysfunction in remote vital organs.



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Forms of tissue stress that occur near the junctions of open and closed lung units.

Figure 29-7



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Stress fracture of the alveolar capillary membrane resulting from ventilator-induced lung injury in a previously normal rat lung. Similar tears have also been demonstrated to occur in human patients with ARDS.

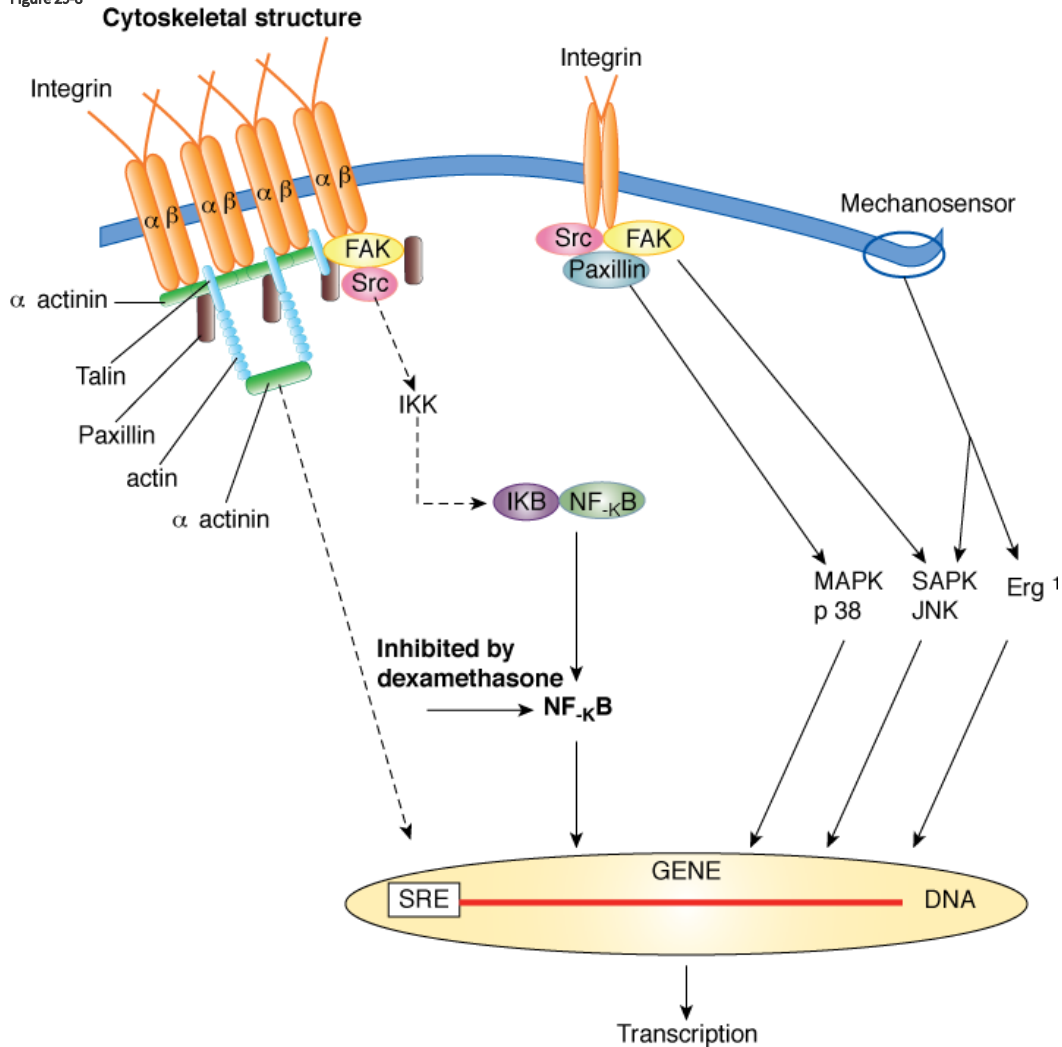
Pathophysiology and Prevention of Ventilator-Induced Lung Injury

From available data, VILI appears to be a complex process initiated by the repetitive application of excessive, stress/strain to the lung's fibroskeleton, microvasculature, terminal airways, and delicate juxtaalveolar tissues. Defining the linkage between stress, strain, and diffuse alveolar damage is currently a subject of intense investigation.^{22,37,38,42,43,45} It seems undeniable, however, that high levels of mechanical stress disrupt the normal functioning of cells that populate the pulmonary microenvironment and that sufficient dimensional strain triggers the release of inflammatory mediators and destructive enzymes (Fig. 29-8).^{43,46,47} Under moderate degrees of strain, such mechanosignaling may be the primary pathway to injury. When the applied mechanical stress is very high, fibroelastic structural integrity may be directly breached, with the inflammatory process a consequence rather than initiator of the observed histopathology. These high tissue-rupturing forces are especially likely to be generated in dependent lung regions, where unstable alveoli are most prevalent (Fig. 29-9). Another important mechanism in the causation of lung tissue damage

airways as they remain closed as airway pressures build or open and reclose with each tidal cycle.³⁶ Damaging

shear forces (those that run tangential to the structure) appear to rip the epithelium from its attachments. The rate at which these tissues expand, conditioned by the inspiratory flow profile, may also play an important role.⁴⁸ Although relatively few studies have focused on this aspect, such a contribution seems likely, as the expansion velocity of the parenchyma is amplified by the reduced number of functioning airways that must accept what is typically an elevated total minute ventilation.

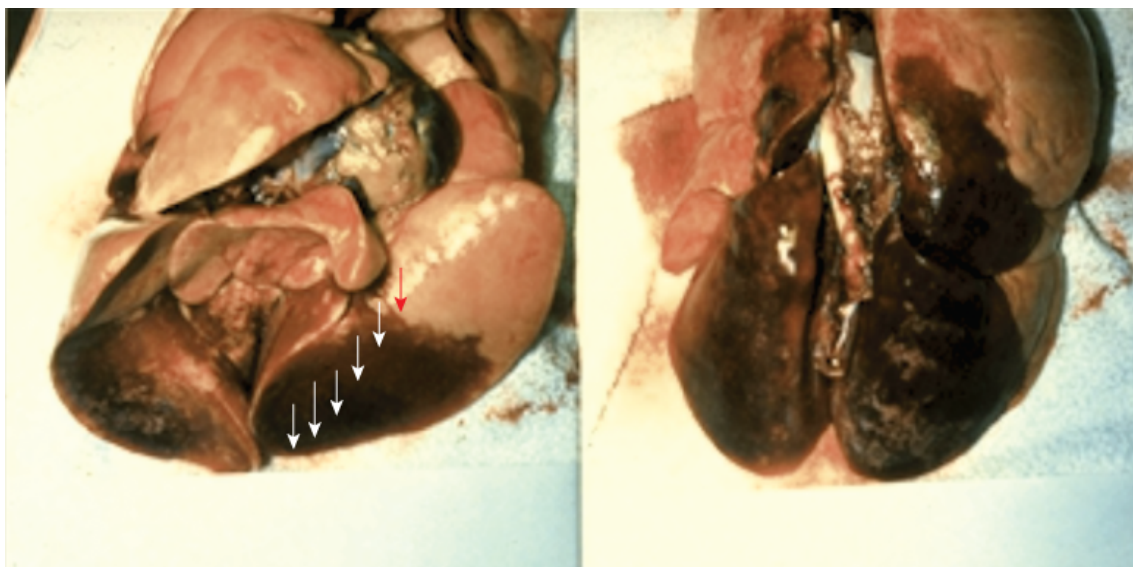
Figure 29-8



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Mechanosignaling pathways of inflammation under conditions of excessive tissue strain. (Used, with permission, from Dos Santos and Slutsky.³⁷)

Figure 29-9



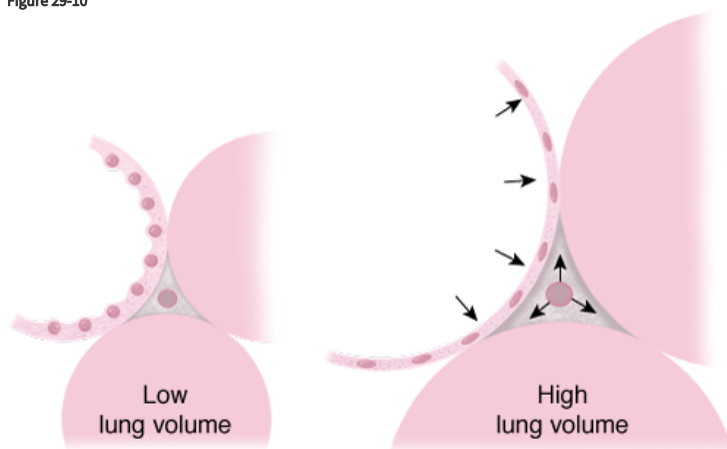
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Excised heart-lung block from a previously normal animal subjected to high inflation pressure and low levels of positive end-expiratory pressure (PEEP). Specimen is shown inflated at 20 cm H₂O airway pressure in the supine (*left panel*) and prone (*right panel*) conditions. Note the sharp demarcation of hemorrhagic edema from relatively normal appearing lung. Damage may progress sequentially from dependent to nondependent regions, as indicated by the sequence of arrows.

From an engineering perspective, mechanical *stress* is a function of transstructural tension; *strain* is the dimension-altering consequence of high transstructural pressure, conditioned by the elastance of the element in question. Although by no means the only determinant of tissue forces, the measurable analog of the stress across the entire lung is transpulmonary pressure—crudely estimated as the difference between static airway pressure (the “plateau” pressure) and average pleural pressure (often estimated by use of an esophageal balloon).⁴⁹ Regional transpulmonary pressures vary considerably, because of the influence of gravity, chest wall irregularities, intraabdominal pressure, mediastinal weight, and vascular-filling pressure (see Fig. 29-2).^{50,51} Available laboratory data illustrate that pleural pressure, however measured, does not accurately reflect interstitial pressure when the lung is edematous and inflamed.^{51,52} Even directional changes of pleural and interstitial pressures may not track together in a nonuniform environment (Fig. 29-10). Although not directly measurable, strain correlates with aerated volume as a fraction of aeratable capacity. As already noted, excessive strain may be experienced when VT exceeds the aerated functional residual capacity of a well-recruited lung by a factor of 2.0⁴⁰ (Fig. 29-11).

Figure 29-10

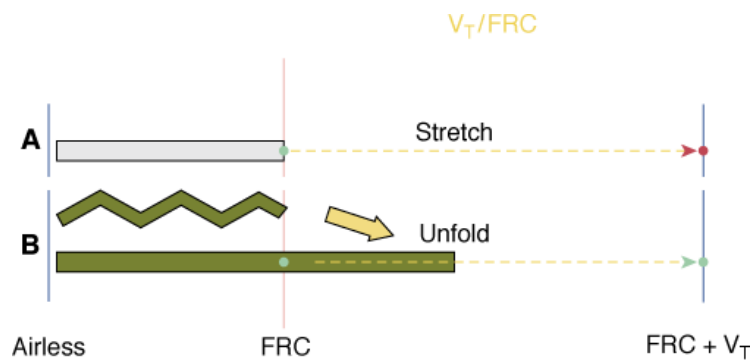


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Mechanical behavior of capillaries embedded in the wall of inflatable alveoli and those microvessels within the interstitial spaces. Wall-embedded capillaries are compressed, whereas extraalveolar microvessels (and perhaps corner vessels) are dilated by lung expansion.

Figure 29-11



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Concept of strain as relates to tidal volume (V_T) and functional residual capacity (FRC). Assuming that the lung has few recruitable units at rest (A), doubling of FRC by the applied V_T will result in excessive tissue strain ($V_T/FRC \geq 2$). Peak transalveolar pressure would reflect that damaging stress. If tidal recruitment occurs during delivery of the V_T (B), the recruited tissue will relieve the stress even though the calculated ratio of V_T to FRC initially is the same. Peak transalveolar pressure would not rise as high as in situation A, but depending on driving pressure amplitude, tidal recruitment might have potential to injure secondary to stress amplification.

How Does Tidal Volume Relate to Tissue Stress?

Once this latter concept is understood, it becomes clear that tissue stress is not a predictable function of tidal volume but rather is strongly influenced by tidal volume in relation to the size (and compliance) of the compartment forced to accept it. Logically, the developed transpulmonary pressure would seem a variable that integrates these relationships. Even this, however, is not the full story—stress amplification and flow-influenced shearing forces at the boundaries of tissues with differing capacities to expand must be considered. Peak tidal transalveolar pressure interacts with end-expiratory transalveolar pressure to determine measurable stress, which varies markedly from site to site across the damaged lung. The alveolar pressures resulting from tidal volume and PEEP are both interactive elements in the injury process, but even when considered together cannot precisely estimate tissue stresses, as interstitial pressure is an unmeasured variable that differs throughout.

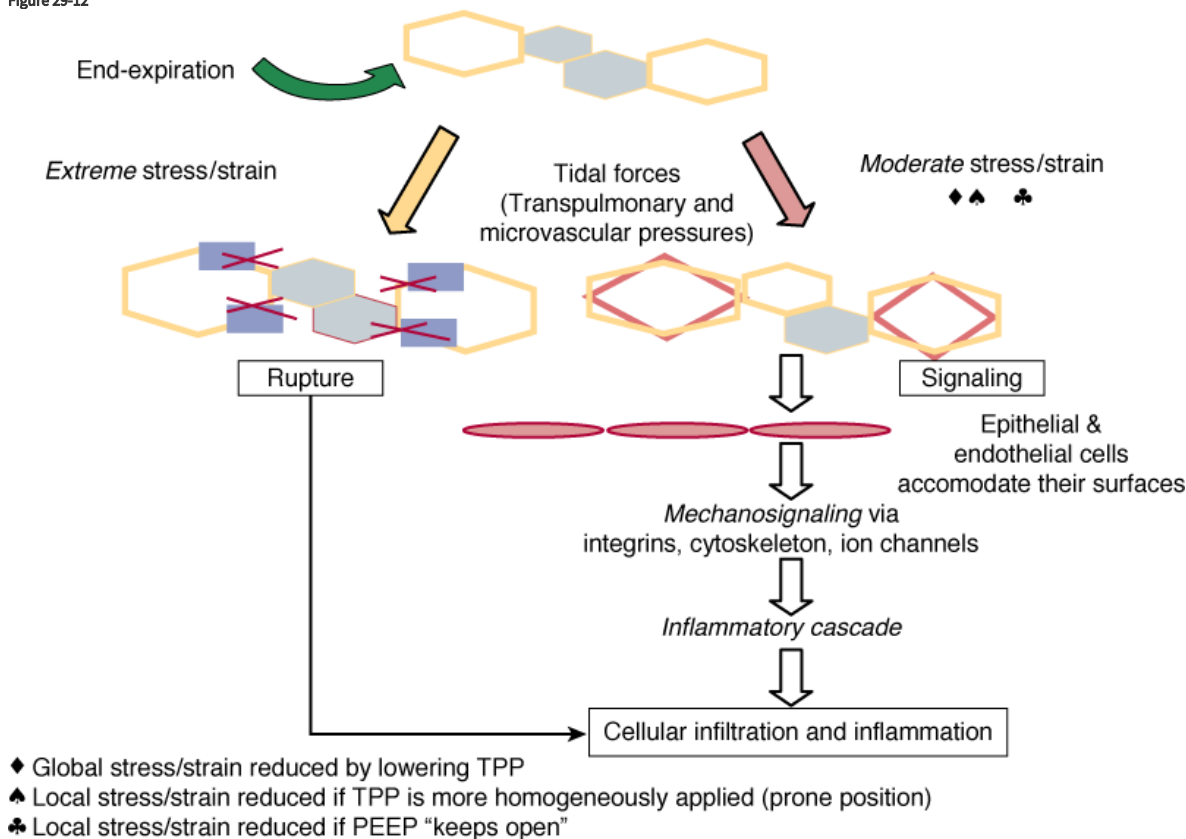
Moreover, the damaging *strain* that results from a given stress depends on tissue fragility. In the setting of ARDS, tissue integrity is likely to be at least normally delicate or even compromised, especially in its earliest phase. The low compliance typically associated with ALI results primarily from the reduced number of aeratable lung units—not from increased stiffness of the individual lung units themselves.^{26,52} The tidal volumes pulled during exercise by a healthy athlete may exceed 25 to 30 mL/kg of lean body weight, whereas much smaller breaths eventually cause damage or lung rupture in the setting of ALI. Although increasing the tidal volume delivered to the same individual will invariably cause tissue stresses to rise in some lung units, a raised tidal volume may actually cause the stresses in other units to *fall*. This apparent paradox owes to the recruiting effect of the higher pressures and perhaps improved ventilation of marginally ventilated units predisposed to absorption collapse.⁵³ At any clinically selected tidal volume, tissue stresses may be reduced or accentuated by the application of PEEP, depending on whether PEEP effectively recruits vulnerable tissue or simply raises plateau pressure. Quests for a universally applicable predetermined tidal volume or PEEP value applicable to all patients would seem destined to be a fruitless exercise.

Modifying the characteristics of the chest wall (e.g., by prone positioning^{54,55}) is an effective mechanism for altering the regional distribution and amplitude of transpulmonary pressure. Even within the same small region, inflationary stresses can vary markedly in magnitude and even in direction between structures situated within microns of one another. Shearing forces, one of the varied forms of mechanical stress resulting from lung inflation, intensify at the junctions of tissues with different compliance values and anchoring attachments.²³ Minimizing or eliminating these irregularities reduces the potential for adverse “stress focusing” and tissue strain. In such a microenvironment, limiting end-tidal alveolar pressure assumes major importance for two primary reasons: (a) A high plateau pressure may overstretch open alveoli, and (b) perhaps more importantly, because junctional tension rises in nonlinear proportion to airway pressure,²³ the plateau pressure acts as a potent lever arm at stress focus points.

By reducing the number of junctional interfaces and by preventing the repetitive opening of terminal lung units at high pressure, recruitment of lung tissue—defined as the sustained reversal of atelectasis on whatever scale it occurs—may be lung protective. When nearly all potentially recruitable tissue is aerated, the lung is said to be “open.”⁵⁶ A given transpulmonary pressure applied to a fully open lung should be associated with less stress than the same pressure applied to a lung with closed units juxtaposed to open ones (Fig. 29-12). Some authors argue that the injured lung should be fully “opened” so as to reduce the potential for repetitive opening and reclosure.⁵⁶ Presently, however, it is not clear that this should always be given highest priority; to what extent repetitive opening and closure of small airways produces injury and whether the prevention of such behavior is the Loading [Contrib]/a11y/accessibility-menu.js, especially when peak tidal pressures are restrained; modest airway pressures may not inflict shearing injury

when opening occurs at low pressure. In addition, because the difference between opening and closing pressures is often quite narrow in units that do open at higher pressure, some degree of tidal recruitment may be unavoidable. Therefore, how much of the lung should be “opened” and what pressure cost is acceptable remain key unresolved questions.

Figure 29-12



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Hypothetical relationship between tissue damage and the severity of mechanical stress or strain during the tidal cycle. Moderate forces applied repeatedly to junctional tissues may result in mechanosignaling of inflammation. Extreme stress or strain causes micro wounds to develop, with inflammation occurring as an epiphenomenon. *PEEP*, positive end-expiratory pressure; *TPP*, trans-pulmonary pressure. (Used, with permission, from Marini JJ, Gattinoni L. Ventilatory management of ARDS: a consensus of two. *Crit Care Med*. 2004;32(1):250–255.)

Pathogenic Cofactors of Ventilator-Induced Lung Injury

Experimentally, a variety of cofactors apart from end-inspiratory, end-expiratory, and driving tidal transpulmonary pressures are important in the generation or prevention of VILI (Table 29-2). Prone positioning confers a protective advantage in both normal and preinjured animals.^{57,58} A potential role for high inspired oxygen fraction as a contributor to iatrogenic injury has been intimated for many years but never convincingly documented in the clinical setting.^{1,37} As other examples, higher precapillary⁵⁹ and lower postcapillary⁶⁰ vascular pressures intensify the injury caused by a fixed ventilatory pattern. As the lung expands, alveolar and extra-alveolar microvessels are compressed and dilated, respectively (see Fig. 29-10). At some intermediate point on the luminal pathway that links them, the pushes and pulls are oppositely directed, giving rise to forces that stress the microvascular endothelium. Energy dissipation across the waterfall created by intermittent zone 2 conditions, vascular interdependence, and opening and closure of the microvascular endothelium could potentially explain the damaging influence of reduced postalveolar vascular pressure.

Table 29-2: Clinically Modifiable Cofactors Influence the Expression of Ventilator-Induced Lung Injury

Vascular pressure
High inflow or low outflow pressures
Ventilation frequency and/or minute ventilation
Hypercapnia may be protective
Position
Body temperature
Inspired oxygen percentage
Pharmacologic interventions

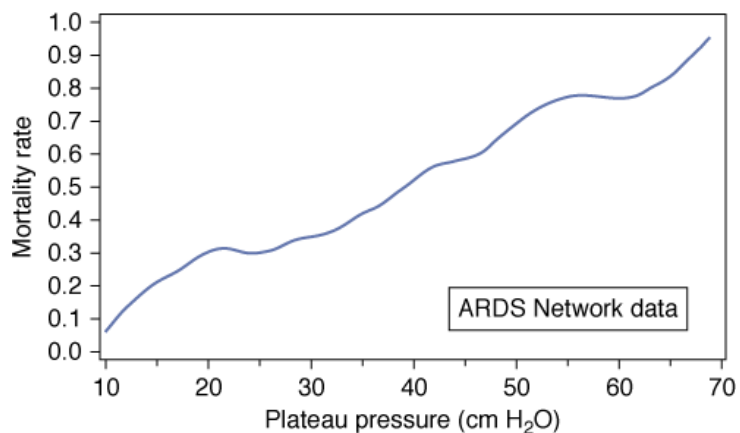
For identical tidal inflation and end-expiratory pressures, reducing respiratory frequency and minute ventilation attenuates or delays damage, provided that the tidal ventilatory stresses are sufficiently high.⁶¹ Large tidal breaths and sighs occur infrequently as part of an inherent pattern of normal breathing. Sighs, for example, are inherent to the natural breathing pattern and are not injurious.⁶² Multiple high-tension tidal cycles are required to signal inflammation. Experimental evidence also demonstrates that effective repair of minor defects within the cell membrane may occur within seconds of reducing stress,^{63,64} and reducing respiratory frequency may allow these repair processes sufficient time to run to completion. Alternatively, less cumulative damage to the lung may occur per unit time as injurious forces unzip structural elements of the lung’s fibroskeleton.

What level of transpulmonary pressure is likely to be damaging, therefore, depends on variables other than the tidal “plateau” pressure. Moreover, when the lung is comprised of large numbers of recruitable units, PEEP attenuates the tendency for high plateau pressures or tidal volumes to cause injury.^{65–67} Consequently, it is difficult to specify an exact level of transpulmonary pressure that serves as an appropriate threshold criterion for safety. From a theoretical standpoint, repeated tidal application of a transpulmonary pressure of 15 to 20 cm H₂O (corresponding in a patient with a normal chest wall to a plateau pressure that may be in the range of 25 to 35 cm H₂O) gives cause for concern, as some higher compliance regions of the injured lung may approach their elastic limits at this pressure. It is worth noting that a transpulmonary pressure of only 15 cm H₂O subjects the normal lung to about two-thirds of its total capacity and is associated with a tidal volume exceeding 2500 mL.¹³

Experimental data suggest that lung tissues can be injured by mechanical stretching forces more easily in the setting of other noxious influences. In other words, it takes a higher stretching force to cause injury in a previously healthy lung than in one that has been exposed previously or concurrently to another inflammatory stimulus, such as endotoxin, hyperoxia, or cytotoxic drugs.^{68,69} Such observations have given rise to the “two-hit hypothesis” for VILI, and underscore the potential vulnerability of the preinjured lung to imprudent ventilatory prescriptions. At the same time, such vulnerability can be viewed as an opportunity to modulate the severity of VILI by altering both nonventilatory and ventilatory factors.

A post hoc analysis of the ARDS Network tidal volume trial (ARMA) indicates that observed mortality paralleled plateau pressure to very low values—considerably lower than 20 cm H₂O (Fig. 29-13).⁷⁰ This correlation, which is difficult to explain entirely by indices of disease severity, suggests the possibility that for the injured lung there is no obvious safety threshold below which ventilator-associated lung damage does not occur. Other data suggest danger at relatively low plateau pressures,⁷¹ but detailed analysis of human clinical trials argue that plateau pressures less than 30 cm H₂O have little impact on mortality once PEEP and disease severity are accounted for.⁷² (Tidal volume must correlate with plateau pressure in any given individual; prestudy tidal volume and plateau pressure data from the ARMA trial, however, showed remarkably poor correlation because compliance [the inverse of elastance] is the missing parameter [Fig. 29-14].)

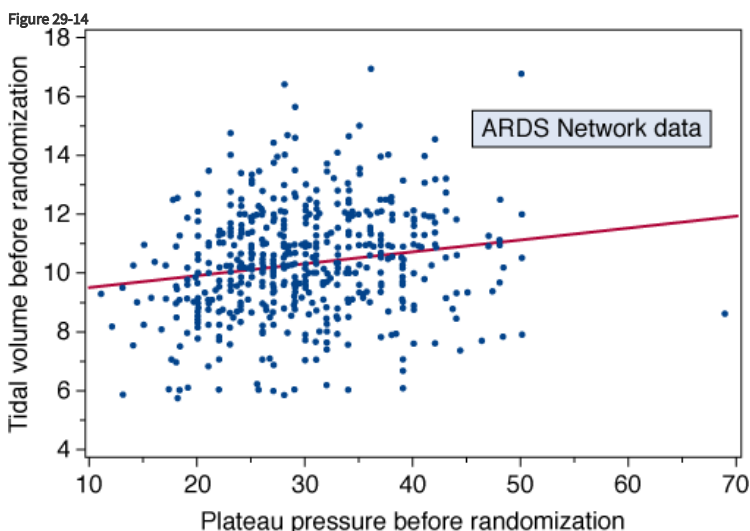
Figure 29-13



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Relationship of mortality rate to plateau pressure generated during the National Institutes of Health-sponsored ARDS Network multicenter trial of high versus low tidal volumes. Note the quasilinear relationship of mortality rate to plateau pressure. (Used, with permission, from Brower et al.⁷⁰)



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Scatterplot of tidal volume versus plateau pressure before randomization to the limbs of the National Institutes of Health-ARDS Network ARMA trial. (Used, with permission, from Brower et al.⁷⁰)

Actions of Positive End-Expiratory Pressure

Elevating the pressure baseline from which breaths are taken or delivered raises mean and end-expiratory lung volumes. Doing so nearly always improves oxygenation to some extent—primarily a function of keeping the lung open. When the breaths are drawn spontaneously or with pressure support, it is customary to call the end-expiratory pressure *continuous positive airway pressure* (CPAP); during breaths of predetermined length, the term is *PEEP*. Both are instrumental in the supportive care of patients with ARDS.

Gas Exchange.

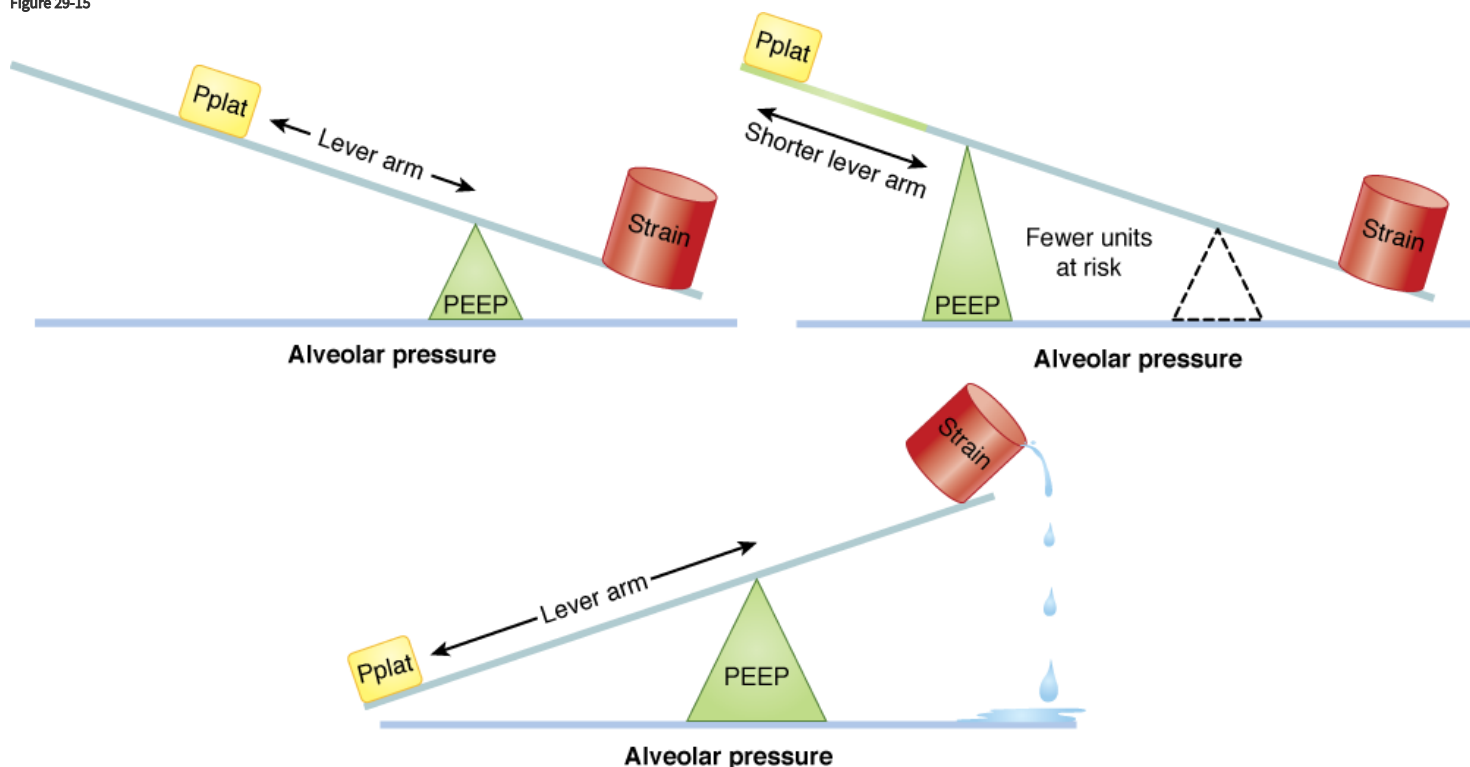
The potential clinical utility of PEEP in improving oxygenation was mentioned in the paper that first brought ARDS to widespread clinical attention.¹ Subsequent work demonstrated its potential to increase ventilation dead space and impair cardiac output by impeding venous return and increased right ventricular afterload.⁷³ In patients in whom PEEP stabilizes lung units that are susceptible to collapse, the response to increasing PEEP is generally to improve pulmonary oxygen exchange efficiency. Many patients show limited or negligible response, however, presumably because the recruitable population of lung units under baseline conditions is small. Infrequently, increasing PEEP can actually cause the partial pressure of arterial oxygen (PaO₂) to fall, presumably by redirecting pulmonary blood flow, by causing pulmonary arterial pressure to rise high enough to shunt venous blood through a patent foramen ovale, or by causing sufficient reduction in oxygen delivery to force an increase in systemic O₂ extraction and

The benefit of PEEP on oxygenation depends on improving functional residual capacity. When patients labor to breathe, as during hyperpnea, expiratory efforts made against PEEP may force the lung's volume to fall below its equilibrium value, effectively storing elastic energy to aid inspiration but interfering with improvement in O_2 exchange. Under these circumstances, eliminating forceful expiratory muscle action (as by sedation) tends to improve O_2 exchange efficiency.⁷⁴

Alteration of Tissue Stress.

Because PEEP has the potential to maintain recruitment of unstable lung units (thereby reducing “stress amplification”), there has been intense interest in its role as a core element of a lung protective ventilating strategy. When plateau pressure is held constant, raising PEEP not only reduces the number of closed lung units, but also shortens the lever arm (“driving pressure”) applied to the unstable lung units at risk to open forcefully (Fig. 29-15). Yet, when tidal driving pressure is preserved, PEEP raises both mean and peak tidal pressures, distends lung units that are already open, redirects blood flow, and alters cardiac loading conditions. Moreover, those lung units at the interface of closed and open units that remain unopened or continue to undergo repeated cycles of tidal recruitment despite an increase of PEEP are subjected to any PEEP-related elevation of end-inspiratory pressure, increasing the tendency for damage to those specific units. Thus, PEEP has the clear potential for benefit or harm depending on the balance among its multiple effects. Prone positioning tends to even the distribution of ventilation and reduce the gradient of transpulmonary pressure across the lung that exists in the supine position.^{55,75} Airway and lymphatic drainage also tend to improve. This improved uniformity facilitates the selection of a single combination of PEEP and tidal volume that achieves a protective strategy for the entire organ.

Figure 29-15



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Hypothetical relationship among plateau pressure PEEP and tissue strain. In the generation tissue strain, the driving pressure for tidal volume (the difference between plateau pressure and PEEP) acts as a force lever arm, whereas PEEP is the fulcrum. *Upper Left.* Levels of plateau pressure (P_{plat}) and PEEP produce strain within acceptable limits. *Bottom Center.* Increasing P_{plat} while keeping PEEP at a level insufficient to hold open unstable units lengthens the lever arm and produces excessive tissue strain. *Upper Right.* When PEEP is increased, the same high P_{plat} that caused damage previously is better tolerated in that fewer lung units are placed at risk after recruitment and the lever arm of driving pressure is shortened.

Recruiting Maneuvers

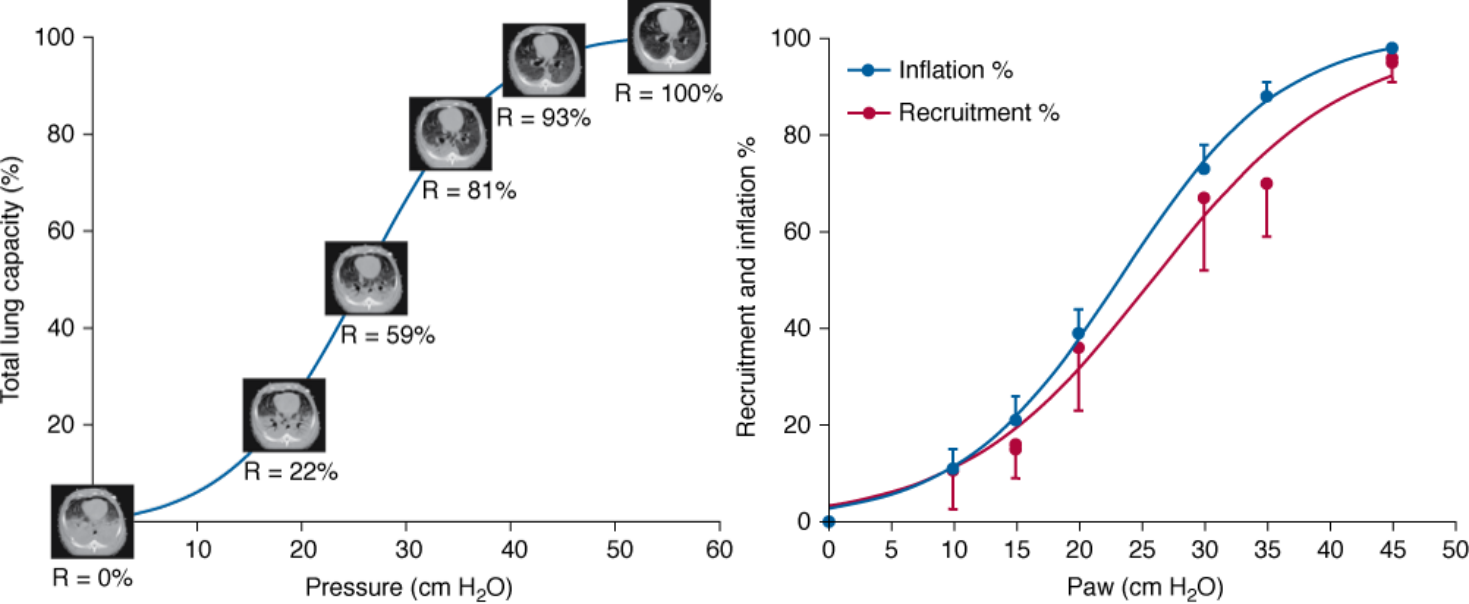
A major clinical challenge is to apply sufficient pressure to keep the lung fully recruited without either increasing the stress applied to tissue that remains closed or overdistingending alveoli that remain patent. Certain steps can be taken to minimize the number of unstable units by reversing those conditions that predispose to compressive and absorptive atelectasis (Table 29-3). PEEP cannot keep open lung units that were never open, and the pressures are reached that considerably exceed the total capacity of the healthy lung (Fig. 29-16). Although

most lung units open at pressures less than 25 cm H₂O, some refractory units of the acutely injured lung may require much higher pressures to establish patency. To reach the “yield” (opening) pressures of refractory lung units requires the initial application of pressures that would be hazardous during tidal ventilation.^{15,27,76} Newly opened lung units tend to close at lower pressures, allowing ventilation with the same tidal volume and PEEP to occur in the context of a more open lung with fewer lung units at risk for opening and closure (Fig. 29-17).

Table 29-3: How Is the Injured Lung Best Recruited?

Prone position
Adequate positive end-expiratory pressure
Adequate tidal volume (and/or sighs?)
Recruiting maneuvers
Minimize edema (?)
Lowest acceptable FiO_2 (?)
Spontaneous efforts (?)

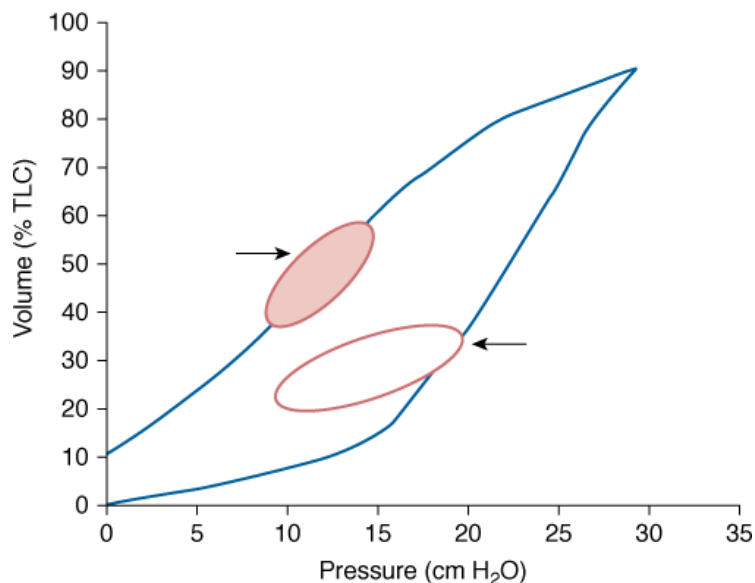
Figure 29-16



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Left. CT scan slices obtained after oleic acid injury of the lung superimposed on the pressure-volume relationship of the respiratory system. Consolidation and radiographic density are greatest in the dependent lung regions and high pressures must be applied, even in this “highly recruitable” lung model, to fully reverse the radiographic evidence for collapse. Percentages denote aerated recruitable tissue. Right. Recruitment and inflation percentages as functions of static airway pressure. In these five patients with ARDS, the pressure–recruitment curve parallels the pressure–inflation curve when both are expressed as percentages of their maximum ranges. (Used, with permission, from Pelosi et al²⁷ and Crotti et al.¹⁵)

Figure 29-17

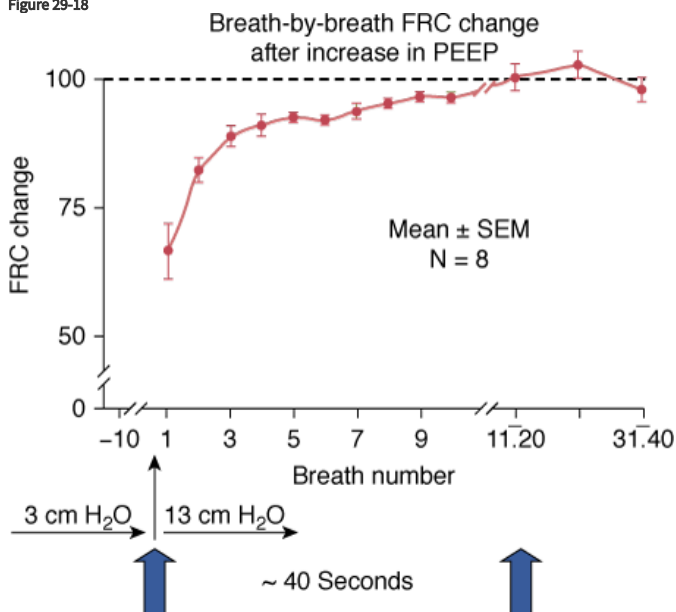


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Effect on the tidal pressure volume relationship (*small loops*) before and after a sustained inflation recruiting maneuver. Hypothetically, first opening the lung by increasing pressure to high values allows tidal ventilation to occur at similar or lower pressures with more lung units open at end-expiration. Note that any volume gained is quickly lost after recruitment if original PEEP and position remain the same.

Because of viscoelastance and other time-dependent force-distributing phenomena, the tendency of a previously collapsed airway to open (or “yield”) is a jointly hyperbolic function of both transmural pressure and time.⁷⁶ Of these, pressure is the most important variable. A variety of techniques, known as recruiting maneuvers, are designed to “open” the lung so that safe and effective combinations of PEEP and tidal volume can be utilized. Each recognizes that recruitment depends not only on the magnitude of transpulmonary pressure, but also on the duration of its application (Fig. 29-18). To consolidate benefit after a successful “recruiting maneuver,” end-expiratory pressure must remain high enough to hold open these newly recruited units once safe tidal plateau pressures are resumed. With few exceptions, this “stabilizing” value of PEEP is higher than the initial one before recruitment.^{77–80} Any benefit from a recruiting maneuver is extremely short lived if PEEP returns to its original value.⁸⁰

Figure 29-18



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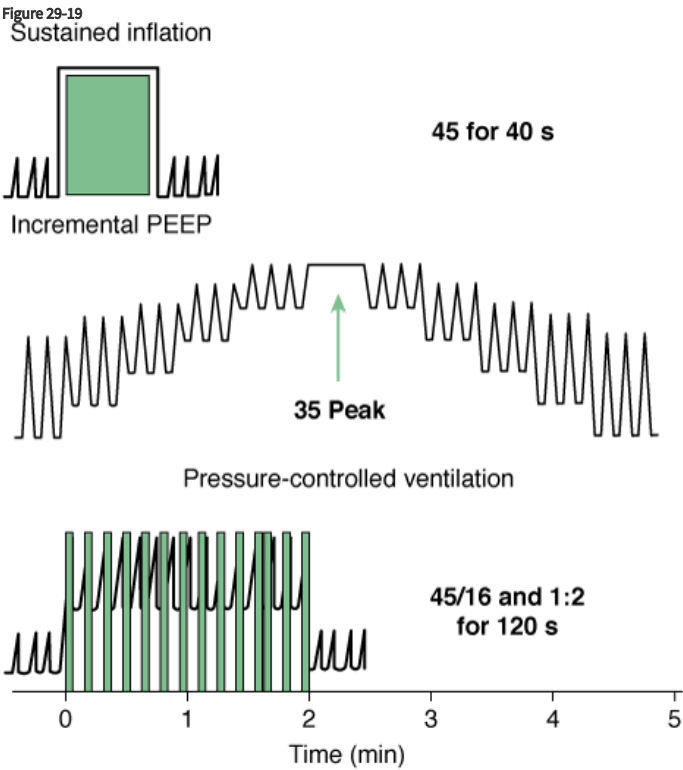
Time course of functional residual capacity (*FRC*) changes in respiratory failure following a step increase of PEEP from 3 to 13 cm H₂O. Multiple tidal cycles are required to fully achieve the ultimate increase in an expiratory lung volume. Although tidal volume would affect the time needed to equilibrate, in this instance, approximately 40 seconds were required. (Used, with permission, from Katz JA, Ozanne GM, Zinn SE, Fairley HB. Time course and mechanisms of lung volume increase with PEEP in acute pulmonary failure. *Anesthesiology*. 1981;54(1):9–16.)

Although most unstable lung units can be kept open with end-expiratory alveolar pressures (total PEEP) of approximately 10 cm H₂O (assuming a normally compliant chest wall), some units close at pressures considerably higher than those that are safe to apply with each tidal breath (see Fig. 29-5).^{15,27} It is a fallacy to consider all injured tissue as potentially recruitable. Unlike most experimental models of ALI,^{27,80} only a small fraction of the lungs of pneumonia-caused (“primary”) ARDS, for example, can be “opened.”^{80–82}

Although a variety of recruiting maneuvers have been described, their efficacy depends on numerous factors (Table 29-4). Moreover, even when indicated, the best technique with which to perform a recruitment maneuver is currently unknown and may well vary with specific circumstances and repeated recruiting maneuvers may be required for maximum benefit. The two most commonly used are the sustained-inflation and the incremental-PEEP and/or fixed driving-pressure methods (Fig. 29-19).⁸¹ The latter is often applied by PCV. When sustained pressure is applied without relief, mean and peak airway pressures are equivalent. This imposes extraordinary backpressure to venous return and poses a high afterload to the right ventricle for the period of its application. Both in experimental models and the clinical setting, ARDS caused by pneumonia (a “primary” form of alveolar damage) appears to be the condition with greatest risk for hypotension and least responsiveness to the sustained-inflation recruiting maneuver.⁸³

Table 29-4: Determinants of Recruitment Maneuver Effectiveness

ARDS category
Inherent <i>potential</i> for response
ARDS stage
Responsiveness diminishes over time
Starting PEEP and tidal volume
Was the lung well recruited to start with?
How much higher than the tidal plateau is the recruiting pressure?
How many more units can be opened?
Postrecruitment PEEP
Duration of response
Aggressiveness and type of recruiting method
Often limited by tolerance



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Three types of recruiting maneuver that combine high pressures with extended application time. Recruitment that employs tidal ventilation with pressure control (pressure-controlled ventilation [PCV]) achieves the same peak pressure for the same cumulative time as sustained inflation but at considerably lower mean airway pressure. PCV also provides multiple recruiting cycles and ventilates during the maneuver. (Used, with permission, from Lim S-C et al.⁸¹)

Sustaining high pressure is believed to be an important component of the recruiting process, whereas the length of time required for its effect remains unsettled. Moreover, it is possible that for the same maximum pressure, briefer applications more frequently may be as effective as fewer cycles applied for a longer time. Recruitment tends to maximize early, while hemodynamic compromise continues to worsen as the high airway pressure is sustained. Mean airway pressure can be considerably reduced while maintaining the same peak airway pressure value—the actual recruiting pressure—using a limited period of tidal ventilation with a constant driving pressure (e.g., <1 minute of pressure control with driving pressure 30 cm H₂O and high PEEP [e.g., 25 cm H₂O]). Thus, the pressure control recruiting maneuver may hold an advantage if pressures beyond those tolerated during the sustained high-pressure method are needed to complete lung opening.^{81,83}

Spontaneous Ventilatory Efforts

Preservation of spontaneous breathing efforts during assisted ventilation may help to improve ventilation–perfusion matching by preferentially ventilating the peridiaphragmatic regions that receive disproportionate blood flow.^{84–86} For the same minute ventilation, varying the amplitude of tidal breaths may also be helpful, either because of better recruitment or improved distribution of blood flow during biologically variable ventilation or “noisy” pressure support.⁸⁶ Whether this redistribution of ventilation and/or perfusion aids *sustained* recruitment, and thus whether it reduces or augments the tendency for VILI, currently remains unknown. The answer may well vary with the intensity of the breathing efforts and the vigor of expiratory muscle activity. It is seldom prudent to preserve spontaneous breathing at the expense of high exertion.

Ventilating Objectives and Decisions in Acute Respiratory Distress Syndrome

The foregoing discussion suggests that a rational lung-protective ventilator strategy should include the following elements: (a) avoidance of high tidal end-inspiratory alveolar pressures; (b) provision of adequate end expiratory pressure to avoid extensive end-expiratory tidal collapse and damaging driving pressures; (c) reduction of regional nonuniformity of mechanical properties; and (d) to the extent possible, avoidance of cofactors that abet the development of VILI, such as high inspired concentrations of oxygen and increased metabolic demands for ventilation and cardiac output. Choice of the appropriate therapeutic targets is fundamentally linked to lowering the risk for iatrogenic lung injury.

Therapeutic Targets

There is little evolutionary precedent for surviving ARDS that requires advanced levels of life support. Yet, the limits of tolerance to abnormalities of gas exchange in this setting and the capacity for the patient to adapt to abnormal blood gases have not been extensively explored and remain largely unknown. Trade-offs must be made when intervening to sustain life, as the supports applied to maintain oxygenation and ventilation—inspired oxygen and positive pressure—are both potentially injurious to the lung.^{68,69,87} The discipline of intensive care evolved as an extension of postoperative care. Therefore, it was natural to employ ventilatory parameters that serve well in that setting, where the general aim is to hold arterial blood gases reasonably close to their preoperative values and large tidal volumes are needed to forestall atelectasis and to avert dyspnea. Three decades ago, awareness of oxygen toxicity was well entrenched, and the acutely injured lung was viewed on the basis of the plain frontal chest radiographs as mechanically uniform—diffusely stiff and therefore tolerant of the high ventilating pressures needed to maintain normal blood gases.

These assumptions were eventually challenged. Premature infants ventilated for infantile respiratory distress syndrome were clearly vulnerable to the application of high ventilating pressure. In adults, the high mortality rate of survivable diseases, such as asthma,⁸⁸ the high incidence of radiographically evident barotrauma,⁸⁹ and newfound awareness of dynamic hyperinflation^{90,91} brought the wisdom of traditional practices and therapeutic targets into question. Reducing ventilatory requirements by allowing partial pressure of arterial carbon dioxide (PaCO₂) to rise was first successfully implemented in the care of patients with status asthmaticus.⁹² Five years later, the same strategy was reported to have been successfully implemented in patients with ARDS.⁹³ Subsequent research has emphasized that acute hypercapnic acidosis occurring in this setting is a complex phenomenon with diverse physiologic effects that include those with potential benefit (e.g., inhibiting inflammation) as well as those with potentially undesirable consequences (e.g., stimulation of ventilatory drive and cerebral vasodilation) (see Chapter 14).⁹⁴ Although guidelines for selecting the most appropriate values for pH and PaCO₂ are elusive and should undoubtedly vary with the individual in question, it is clear from accumulated experience that mild to moderate deviations from the normal ranges of both parameters are generally well tolerated. At present, the dangers of VILI are perceived to outweigh those attributable to either inspired oxygen or hypercapnia.

Important principles gathered from laboratory experiments, shared clinical experience, observational studies, and randomized clinical trials have emerged regarding the application of ventilatory support to lung-injured patients (see [Table 29-1](#)). Noninvasive ventilation has been reported to be successful in patients with mild to moderate severity of injury,^{95–97} and when feasible to employ, benefit is likely to accrue from the need for less sedation and the lower incidence of infection associated with this approach.^{96,97} Although the interface between ventilator and patient continues to improve, only limited pressure can be tolerated, many patients cannot protect the upper airway, and the lung must be kept well recruited to avoid hypoxemia, precluding intermittent removal. For these reasons, airway intubation is virtually always required in cases of severe lung injury. That said, efficient and less-invasive extrapulmonary gas exchangers may complement noninvasive ventilation to make avoidance of invasive ventilation a genuine possibility for the alert and cooperative patient.^{98,99}

Patterns of lung expansion differ for passive and actively assisted breathing. Preservation of muscular effort conceptually holds an advantage regarding the matching of ventilation to perfusion—at least in the healthy lung. The same advantage may hold for the acutely injured lung as well, provided that breathing remains comfortable and unlabored. Vigorous efforts, however, are undesirable for several reasons: increased ventilatory workload, oxygen consumption and CO₂ production, increased cardiac output and pulmonary blood flow (which may accelerate edema formation and possibly VILI), and expiratory muscle contraction that counters the effects of PEEP.¹⁰⁰ Whether gentle or vigorous, it has yet to be convincingly shown that patients who expend effort in breathing experience a lower risk of mortality. In fact, a recently published study demonstrated a potential survival advantage for patients receiving paralytic agents in the early phase of ARDS.¹⁰¹

Modes of Ventilation for Acute Respiratory Distress Syndrome

An extensive array of ventilatory options is available to the clinician for treating patients with ARDS. Each of these methods is described in detail elsewhere in this volume. As a general statement, it is accurate to state that the choice of ventilating mode is of much less importance than how the selected mode is implemented. When guided by the principles of attaining efficient gas exchange, targeting appropriate therapeutic objectives, and protecting the lung against iatrogenic damage, many different selections can be justified. What is presented here is a brief outline of some of the more important characteristics applicable to ALI.

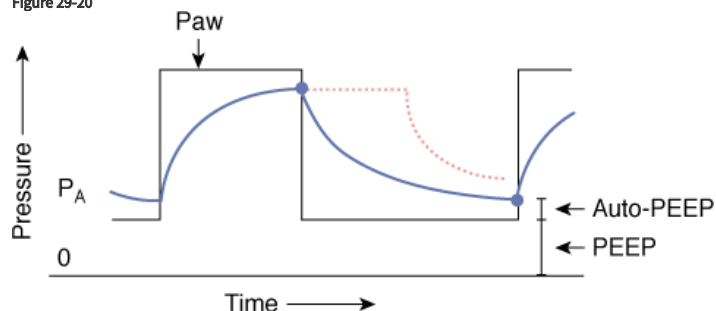
For the past two decades, four modes have served to ventilate patients with ARDS at conventional frequencies: (a) flow-controlled, time-cycled, or volume-cycled ventilation (*volume control*); (b) pressure-targeted, time-cycled ventilation (*pressure control*); (c) pressure-targeted, flow-cycled ventilation (*pressure support*); and (d) combination modes in which a specified number of stereotyped time-limited machine cycles (pressure or volume controlled), which are specified by the clinician and synchronized to be triggered by patient effort, are delivered intermittently at evenly spaced intervals among additional breaths of variable size and depth taken by the patient (synchronized intermittent mandatory ventilation). As described in detail elsewhere in this book, flow and pressure cannot be controlled simultaneously, as the energetics of ventilation require one or the other to become a dependent variable in order to satisfy the equation of motion for ventilating the respiratory system.¹⁰²

Flow control ensures that the desired tidal volume will be delivered reliably but obligates the patient to a specified inspiratory flow contour, independent of flow demand. Therefore, alveolar pressure, a function of delivered volume and compliance during passive ventilation, has the potential to ascend to dangerous levels. The flow profile delivered may be unchanging (“square”) or decelerating. Pressure control offers the flexibility to satisfy flow demand and under passive conditions ensures that alveolar pressure rises no higher than a known peak airway pressure. (During active breathing [assisted cycles], the maximum transpulmonary pressure is not regulated by pressure control.) Flow is inherently decelerating, which may moderately improve distribution of ventilation among heterogeneous lung units, as compared with constant flow. The potentially adverse trade-off with using pressure control is that the delivered volume is a function of the impedance to breathing and any backpressure opposing inspiratory flow, so that tidal volume may change abruptly with muscular activity, airway resistance, lung and chest wall compliance, or auto-PEEP. It is worth noting that with airway pressure capped and the patient passive, the “strain ratio” of tidal volume to functional residual capacity tends to decline as disease worsens, conferring a lung-protective effect not seen with volume-controlled ventilation.¹⁰³

In recent years, the imperative to maintain consistent ventilation and nearly normal levels of PaCO₂ has declined in its relative importance. Awareness of the potential for high ventilating pressures to inflict iatrogenic lung injury, coupled with a growing clinical experience that suggests the safety of higher than normal PaCO₂ values (permissive hypercapnia) have caused many clinicians to adopt pressure control as the default mode. In the absence of effort and provided that both tidal volume and end-inspiratory pause pressure are closely monitored, the choice of pressure or volume control makes little practical difference to physiology or outcome.

Inverse-ratio ventilation, a mode that extends inspiratory time with the intent of raising mean airway pressure and capping applied airway pressure while maintaining adequate ventilation, is typically applied with pressure control.^{104,105} Preventing airway closure at end expiration, however, may require generation of auto-PEEP that narrows the inspiratory pressure difference and limits tidal volume ([Fig. 29-20](#)). Although there may be rare cases in which inverse-ratio ventilation appears not to offer any notable advantage over conventional-ratio ventilation that is applied with adequate PEEP.

Figure 29-20



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Concept of inverse-ratio ventilation using pressure control. As airway pressure is applied and released, alveolar pressure rises and falls exponentially (*heavy line*). Extending the inspiratory time fraction increases mean airway pressure and levels of auto-PEEP, without changing peak alveolar pressure. PA, alveolar pressure; Paw, airway pressure.

During vigorous breathing, as during the first days of severe ARDS, it may be difficult to avoid patient-ventilator dyssynchrony with either assist-control or pressure-control ventilation because both require preset inspiratory times. Pressure limitation and failure to deliver full tidal volumes may then occur. In these instances, high-level pressure support, a flow-cycled mode, offers response flexibility. When the chest is very stiff, however, pressure rise time should be slowed and flow offswitch trigger minimized so as to avoid overly abrupt and ineffective tidal cycling. When used alone or in combination with intermittent time-limited mandatory machine cycles (synchronized intermittent mandatory ventilation), pressure-support ventilation may minimize the timing collisions that otherwise occur between the cycling phases of patient and machine. In any form of pressure-targeted or flow-regulated breathing, the ventilator's alarms should be carefully set so as to avoid over application of pressure or underventilation.

Newer Modes of Ventilation

For patients capable of sustaining ventilatory effort, it is an option to elevate the airway pressure baseline, allowing the patient to draw breaths from that higher volume level. When airway pressure is maintained relatively constant throughout both phases of the breathing cycle, the condition is termed CPAP.¹⁰⁶ If the baseline shifts periodically with the patient able to draw breaths unimpeded through a valveless system at each level, ventilator assistance is given to an extent governed by the difference between pressure baselines and the frequency with which these baseline shifts occur. This mode is known as *biphasic airway pressure* (BIPAP).¹⁰⁷ When a high CPAP baseline is released only for a very brief period and then quickly restored to its original level, BIPAP contracts into *airway pressure release ventilation* (APRV).^{108,109} The putative advantage of each of these modes for the care of patients with ARDS is that they encourage maintenance of an open lung with their high pressure baselines and ensure that the spontaneous breathing pattern is preserved. Neurally adjusted ventilator assist and proportional-assist ventilation regulate airway pressures in direct accordance with spontaneous respiratory motor output and have been used effectively in some patients with ALI, but their safety and effectiveness for this condition remain to be shown (see [Chapters 12 and 13](#)).^{110,111} As with so many mode options, however, no definitive evidence exists to confirm their relative advantage over a well-adjusted traditional approach with similar mean airway pressure.

High-Frequency Ventilation.

Awareness of the tissue-damaging potential of applying high end-inspiratory plateau pressures with insufficient end-expiratory pressure to keep unstable alveoli open ignited interest in using methods that apply very small tidal volumes so rapidly that alveolar collapse has no time to occur. Over the past three decades, various techniques for providing high-frequency ventilation have been explored with enthusiasm, tried tentatively in the clinical setting, and then reluctantly set aside.^{112,113} These have included positive-pressure ventilation with valved-circuit closure to separate the phases of the ventilatory cycle (as traditionally done) but applied at nonconventional frequencies and tidal volumes. The technical demands of such high-frequency valving have limited its operating frequency range, efficiency, and safety. Another form of high-frequency ventilation, jet ventilation, uses a lung-directed injector pulsating at rapid frequencies into an open circuit. Problems with gas trapping, drying of airway secretions, and limited efficacy have dampened enthusiasm for its use in critically ill patients. At present, a third form, high-frequency oscillation, is in most widespread clinical use for patients with ARDS.^{114–116} This “open circuit” technique uses a rapidly reciprocating piston-like action to rapidly vibrate the air column, building and releasing small alveolar pressure excursions in the process. A fresh-gas source allows effective gas exchange via mechanisms that complement the bulk flow responsible at lower frequencies (see [Chapter 19](#)). Deep sedation is usually required, and mean alveolar pressure is maintained at a high level. Although experimental and clinical data amply document the feasibility of high-frequency oscillation for the setting of ALI in adults, the prevailing opinion is that this unfamiliar and seemingly exotic technique seems to offer little advantage over ventilation performed conventionally with equivalent attention to the principles of lung protection.¹¹⁷

The combination of conventional ventilation with superimposed oscillation of the airway-pressure baseline (vibropercussive ventilation) has been used with the objective of improving secretion elimination and reopening plugged or collapsed airways.¹¹⁸ Here again, the mode has its advocates and adherents, but convincing evidence to substantiate its advantage has not yet appeared.

Adjuncts to Mechanical Ventilation

Attention to limiting tissue strain mandates that the pressure driving each tidal cycle (the difference between end-inspiratory and end-expiratory pressures), as well as the number of high-pressure cycles applied per unit time be kept within acceptable bounds. These requirements limit tidal volume and minute ventilation and stimulate interest in methods that reduce the demands for ventilation and oxygenation, reduce local tissue stresses, and improve the efficiency of pulmonary gas exchange, all without the need for additional ventilating pressures or higher ventilating frequency (Table 29-5). The majority of these “adjunctive” techniques are discussed in detail elsewhere in this book. Here it should be emphasized that perhaps the most effective and universally applicable means for avoiding the need for high ventilator pressures is to reduce the demand for them by the patient or to readjust the clinician’s therapeutic targets for ventilator support. Thus, avoidance of high fever, pain, agitation, and metabolic acidosis reduces oxygen and ventilation demands, as well as the need for cardiac output—an important potential cofactor for VILI in the experimental setting. Optimizing oxygen delivery by improving cardiac performance and providing adequate hemoglobin concentration also minimizes the ventilatory requirements. At present, improved and simplified technology has spurred enthusiasm for using extracorporeal CO₂ removal to reduce ventilator demand and for extracorporeal membrane oxygenation when oxygenation cannot be assured by conventional means. Both were used extensively and effectively during recent epidemics of H1N1 influenza-induced ARDS.¹¹⁹

Table 29-5: Therapeutic Adjuncts to Mechanical Ventilation of Acute Respiratory Distress Syndrome

Minimize O ₂ demands
Optimize O ₂ delivery
Recruiting maneuvers
Prone positioning
Inhaled nitric oxide/inhaled prostacyclin
Tracheal gas insufflation
Corticosteroids and (?) other drugs

Using the prone position evens the distribution of transpulmonary pressure, reducing local tissue strains and effectively recruiting well-perfused dorsal parenchyma, thereby improving oxygenation in most patients.⁵⁵ Methods for improving gas-exchange efficiency include those directed at oxygenation (inhaled nitric oxide^{120–122} or inhaled prostacyclin¹²³), and those that lessen wasted ventilation (tracheal gas insufflation^{124–126} or reduction of apparatus dead space¹²⁶). Although often considered to be a ventilatory adjunct of limited value, recruiting maneuvers are incorporated by many practitioners, including this author, as an entrenched part of clinical practice.¹²⁷ In difficult cases, the recruiting maneuver affords a logical means of reopening the atelectatic lung so that the appropriate values of PEEP and tidal volume may be selected (see the section General Guidelines for Ventilatory Management).

Setting the Ventilator: Recommendations for Practice

Given the joint potential for the ventilator to offer life support or to extend the severity and duration of the illness, the prescribed settings for the ventilatory cycle are of unquestioned importance. How best to achieve the appropriate trade-off, however, remains a topic of active debate. The available clinical database, albeit difficult to reconcile, appears broadly consistent with the highly consistent scientific body of information that addresses ventilator usage in the setting of ALI.

Insights from Clinical Trials of Lung Protection

With the rise of “evidence-based practice,” physicians have sought guidance from clinical trials that have addressed the relative merits of different ventilation strategies.^{2,3,128–130} Generally speaking, these well-intentioned efforts have been compromised by overly broad disease definitions and loose co-intervention controls. Yet some valuable insights have been generated. Studies in which the highest tidal volumes and pressures were applied in the “control arm” have shown benefit from low-tidal-volume ventilation.^{2,3} Results were particularly impressive when higher PEEP was used in conjunction with small tidal volumes in a setting where emphasis was placed on establishing stable recruitment, avoidance of high ventilating interventions consistently.^{2,131}

Only one of the several studies that randomized selectively on tidal volume—by far the largest yet published—successfully demonstrated mortality benefit for a smaller-tidal-volume approach.³ The physiologic impact of tidal volume—mediated through transalveolar pressure—depends on compliance. The ratio of tidal volume to functional residual capacity may crudely approximate tissue strain,⁴⁰ as do the magnitudes of transalveolar pressure and its tidal excursion. In the clinical setting, the former can be readily measured by some ventilators, while the latter is computed by the difference between static airway and esophageal pressures.⁴⁹ Provocative examinations of the data collected in the ARDS Network ARMA study and a meta-analysis of all published trials addressing the selection of tidal volume have suggested that “lower is not necessarily better,” especially when compliance is less severely impaired and periodic sighs or recruiting maneuvers are not employed.^{132,133} Knowing that tidal volume is only indirectly linked to tissue strain, inflammation, and rupture (consider the noninjurious effects of high tidal volumes during exercise), it is interesting to speculate that the recruiting effects of higher tidal volumes might actually have a salutary effect on inflammatory signaling if peak transpulmonary pressure were kept below the “overstretch” signaling threshold and an appropriate level of PEEP were used. Whatever the validity of that controversial argument, the collective results of these clinical studies have focused attention on transalveolar stresses rather than on tidal volume per se. They have also demonstrated that the levels and effects of hypercapnia experienced during low-tidal-volume ventilation, although complex,^{94,134–136} generally are modest and well tolerated.

The National Institutes of Health-sponsored trial of high versus moderate PEEP failed to demonstrate a significant survival advantage for patients allocated to the high PEEP group when both were ventilated using small to moderate tidal volumes.¹³⁷ Important considerations, however, were (a) neither group was exposed to plateau pressures that were clearly in a dangerous range; (b) recruitment potential was not stratified, so that patients who were not likely to benefit were assigned to both groups; and (c) no recruiting maneuver preceded PEEP application, nor was PEEP regulated as in conventional clinical practice. Unfortunately, there was also a failure of the randomization process, so that a disproportionate number of older patients were assigned to the high-PEEP limb. Two subsequent trials of high PEEP and small-to-moderate tidal volumes demonstrated some benefit of a higher PEEP strategy, but no overall mortality benefit.^{138,139} The importance of PEEP is likely to depend on the severity of illness and the recruitability of lung units, which tend to track together, as well as the plateau pressure achieved.^{140,141} A Spanish multicenter trial of modest size appears to confirm that PEEP is an integral part of a lung-protective approach,¹³¹ an implication supported by an analysis of “real-world” ventilator practice and outcomes.⁷² A meta-analysis of high PEEP strategy trials indicated that high PEEP exerts a mortality benefit for those with greatest disease severity.^{142,143}

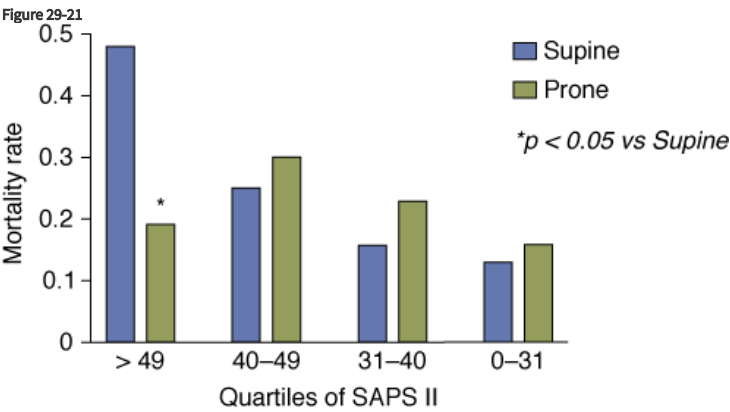
Recruitment maneuvers have also been examined in a clinical trials format as a substudy of the high-PEEP National Institutes of Health investigation.¹⁴⁴ The recruiting maneuvers were applied only in the high-PEEP limb, and the peak pressure achieved was limited to 35 cm H₂O. The small separation between plateau pressure and PEEP, the underlying disease characteristics of the study population, the relatively high level of PEEP at baseline, and the failure to augment PEEP after the recruiting maneuver undoubtedly biased the result against showing a significant benefit. Indeed, a meta-analysis of studies that employed recruiting maneuvers concluded that they have no routine place in ARDS management.¹⁴⁵ Again, however, it must be understood that most studies have used them as a supplement to the ventilator strategy, which remained unmodified, and not as a means for setting PEEP by decremental titration.

Finally, notwithstanding considerable theoretical appeal (Table 29-6), oxygenation benefit,¹⁴⁶ and suggestive experimental evidence,^{57,58,147} four relatively large randomized trials^{148–151} failed to demonstrate a statistically significant mortality benefit for prone positioning in broad samples of patients of varying severity with ARDS. A French study reported a lower incidence of ventilator-associated pneumonia in the prone cohort,¹⁵⁰ but this did not translate into a survival advantage for that trial limb. A large Italian trial utilized prone positioning for less than one-third of each day, and showed no overall mortality benefit.¹⁴⁹ Notably, however, patients who responded by improving CO₂ exchange and those with most severe physiologic disturbances fared better than nonprone patients.¹⁵² Moreover, a prematurely truncated Spanish trial,¹⁵¹ in which prone positioning was maintained for three-quarters of the day, showed an impressive separation (25% relative risk of death) between the supine and prone treatment arms. Unfortunately, neither study entered its targeted number of patients. The large follow-up Italian trial that entered only patients with moderate and severe illness strongly suggested that the most seriously affected patients, those exposed to higher ventilating pressures, and those who demonstrate improved ventilation efficiency are most likely to benefit (Fig. 29-21).^{153,154} As with PEEP, a meta-analysis of prone studies that focused on the most severely ill cohorts convincingly shows the benefit of the prone position in that specific patient category.¹⁵⁵

Table 29-6: Common Effects of Prone Positioning in Early Acute Respiratory Distress Syndrome

More homogeneous transpulmonary pressure
Increased and sustained traction on dorsal lung units
Better $\dot{V}/\dot{Q}$ matching
Tendency for recruitment
Improved airway drainage
Improved lymphatic drainage
Modestly increased FRC
Reduced tidal tissue strain [V_T /FRC (and VILI?)]

Abbreviations: FRC, Functional residual capacity; $\dot{V}/\dot{Q}$, ventilation–perfusion ratio; V_T , tidal volume; VILI, ventilator-induced lung injury.



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Effect of prone positioning on mortality as a function of illness severity. The cohort of most severely ill patients with acute lung injury by Simplified Acute Physiology Score (SAPS) II scoring benefit from proning. Less severely affected patients did not. (Data from first Italian multicenter clinical trial of prone vs. supine positioning in ARDS.¹⁴⁹)

Disappointingly, clinical trials of various other adjuncts to ventilation have also not demonstrated major outcome advantages.^{156–160} From a physiologic perspective, this is hardly surprising. Precise numerical guidelines for selecting PEEP, tidal volume, and body position that are applicable to any given individual patient should not be expected from the results of studies conducted in a diverse sample population whose inclusion criteria are broadly inclusive and management details are unconstrained. In the end, the best that can be hoped for is a proof of principle—not a prescription for care. However internally valid such trials may be, they do not correspond to the “real-world” environment wherein comorbidities impinge and management of potentially influential variables (such as PEEP, PaCO₂, fluid management) is not protocolized. What follows is an approach to the care of the individual patient with ARDS based on an understanding of the physiologic principles that must be brought to bear in the complex clinical environment.

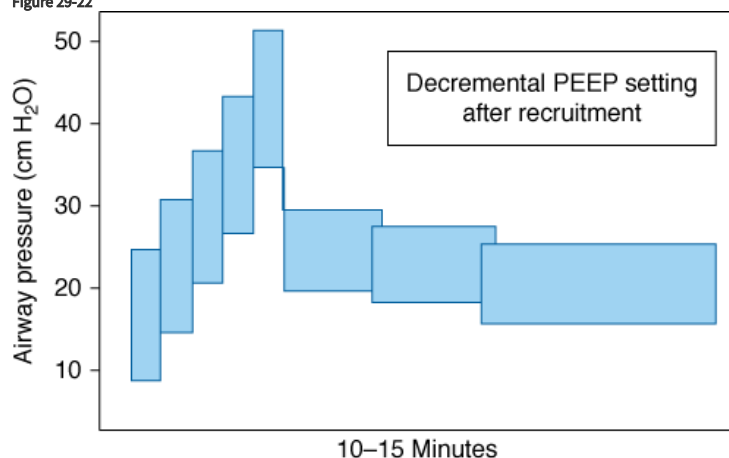
General Guidelines for Ventilatory Management

Key guiding principles include the following. First, adjust ventilator parameters empirically, rather than by formula-driven rules; prioritize patient comfort and safety. Second, assign the prevention of mechanical trauma precedence over maintenance of normocapnia and avoidance of oxygen toxicity. Although no exact upper limits for acceptable plateau pressure or fractional inspired oxygen concentration (F_IO₂) can be specified, very high F_IO₂ pose a risk for absorption atelectasis as well as oxygen toxicity. Therefore, F_IO₂ should be held to less than 0.7 whenever possible. Third, consider the impact of chest wall stiffness (including abdominal contents) on transpulmonary pressure and gas-exchange efficiency. In questionable situations, determine abdominal (bladder) and/or esophageal pressures to help estimate the transalveolar pressure.⁴⁹ Fourth, monitor hemodynamics as well as mechanics and gas exchange when regulating ventilatory therapy. Assess cardiac function by echocardiography and use reliable, noninvasive methods for tracking cardiac output if available. Although certainly not required for all, the much maligned pulmonary artery catheter is still quite useful in patients with complex cardiopulmonary problems.^{161,162} Wide respiratory variation in the arterial pulse pressure, as well as large variations in the venous pulse pressure, suggest the need for additional intravascular

volume.¹⁶³ A surrogate for measuring hemodynamics directly may be to monitor the central venous oxygen saturation. A value greater than 0% and a difference of 25% or less between arterial and mixed venous saturations strongly suggest an adequate cardiac index ($>2.5 \text{ L/m}^2/\text{min}$).¹⁶⁴ Fifth, in severe cases, attempt to minimize ventilatory demands and thereby reduce needs for high airway pressures, high rates of gas flow, and increased cardiac output. Sixth, incorporate the “challenge” principle in making therapeutic decisions, regarding both the intensification and the withdrawal of therapeutic measures. Examples of such challenges include recruiting maneuvers to assess lung-unit instability and closely monitored (and preferably quickly reversible) challenges of fluid administration or removal (e.g., leg raising). Seventh, whereas aggressive fluid resuscitation is mandatory in the first phase of support, attempts should be made to avoid gross volume excess. Diuretic drips, repair of serious albumin deficits, and continuous hemofiltration hold important therapeutic potential once the resuscitation phase has been completed. Eighth, unless otherwise contraindicated, use prone positioning when high values for ventilatory pressure, PEEP, and FI_{O_2} are needed to maintain adequate supine arterial oxygen tension. Ninth, employ pressure limited forms of ventilation (e.g., pressure control, pressure support, or BIPAP/APRV) for evaluation of thoracic mechanics and ongoing management.

As a general rule, the desired goal is to use the least PEEP and tidal volume needed to achieve acceptable gas exchange while avoiding extensive tidal collapse and reopening of unstable lung units. Knowing that moderate hypercapnia generally is well tolerated, therapeutic targeting priorities are directed toward lung protection and maintenance of appropriate hemodynamics and oxygen delivery. Recruiting maneuvers help to characterize PEEP responsiveness, to determine the relative status of intravascular filling and response to altered cardiac loading conditions, and to set the PEEP–tidal volume combination (Fig. 29-22). Prone positioning is strongly considered in severe cases. On rare occasions, noninvasive mechanical ventilation using a comfortable full face mask (or a helmet, if available) may overcome short-lived deficits of oxygen exchange without the need for intubation. In practice, however, the needs to control the airway, to reduce ventilatory requirements, to apply high levels of PEEP, and to sustain support for extended periods usually preclude the use of noninvasive ventilation.

Figure 29-22



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Procedure for setting decremental PEEP. Final PEEP is set after brief recruitment procedure, using progressive steps of PEEP with a fixed driving pressure. Note that PEEP is released abruptly from its highest level to a high tidal value, which is gradually reduced until deterioration of the monitored parameter (compliance or oxygenation) becomes apparent. After re-recruitment, tidal PEEP would then be set to the next higher step.

In the first phase of ventilatory support, patient comfort must be assured and ventilatory effort kept to a minimum. Existing data strongly suggest that muscle relaxing drugs are effective and well tolerated initially by many. Controlling ventilation for longer than 48 to 72 hours, however, is likely to contribute to muscle wasting and ventilator-induced diaphragmatic dysfunction.¹⁶⁵ Modes such as APRV, BIPAP, and high-frequency oscillation have persuasive advocates and considerable theoretical appeal.^{109,166,167} The existing database and shared personal experience, however, have not provided convincing evidence that they offer a great deal beyond that which can be accomplished with carefully adjusted PCV in a well-sedated patient.

All patients should be assessed for severity of disease and for recruitment potential. After deficits of intravascular volume have been addressed and hemodynamics have been optimized, recruitment potential is gauged by applying high-level PCV: PEEP of 15 to 25 cm H₂O, driving pressure of 30 cm H₂O, and plateau pressure 50 to 60 cm H₂O for a maximum of 1 to 2 minutes, as tolerated. Even higher pressures may be appropriate for some severely affected patients and for patients with very stiff chest walls—for example, burn victims with chest wall edema or eschar. Although sustained inflation with high pressure has been traditionally used, widely employed, and selected for most reported research,¹⁶⁸ it is no more effective and tends to be less well-tolerated hemodynamically than a recruiting method based on PCV that achieves lower average pressure but similar peak pressure during its inspiratory phase.^{72,83} If oxygenation and lung mechanics do not improve substantially with high-level PCV as a recruiting

technique, the patient is considered to have low recruiting potential *in that position and at that specific time*. Management goals in the “recruitable” group emphasize the maintenance of high-level PEEP, whereas in poorly recruitable patients, PEEP is maintained as low as feasible—generally in the range of 7 to 10 cm H₂O. In both groups end-inspiratory plateau pressure is kept less than 30 cm H₂O, except when chest wall compliance is very low. Periodic sighs may be advisable when very low tidal volumes (<5 to 6 mL/kg of predicted weight) are in use.^{169,170}

Adjusting Positive End-Expiratory Pressure

Setting PEEP is an individualized empirical process that involves some degree of compromise (Table 29-7). Patients with an extensive “recruitable” population of lung units should respond to increased PEEP and recruiting maneuvers by demonstrating improved alveolar mechanics and improved gas exchange, reflected both by increased PaO₂ and a reduction in the ration of minute ventilation to PaCO₂. These salutary changes are accompanied by only marginal effects on hemodynamics, as judged by systemic blood pressure and central venous oxygen saturation.¹⁶⁸ Assuming an unchanged rate of CO₂ production, the latter index—like a dead-space calculation—reflects the efficiency of CO₂ elimination, which is expected to improve with recruitment and deteriorate with overdistension. Inspiratory crackles (rales) audible over the dependent zones of the chest suggest that recruitment and derecruitment are occurring with each breath and indicate that recruitment maneuvers and higher levels of PEEP may be indicated to silence them.¹⁷¹ Crackles occurring late in inspiration are of particular concern because they may originate in units opening under relatively high pressures. In gauging response to PEEP, it is important to consider CO₂ exchange as well as oxygenation response. With rare exception (e.g., when PEEP-impaired cardiac output causes mixed-venous O₂ content to fall), PaO₂ tends to increase when PEEP is applied. This oxygenation improvement, however, may be accounted for either by recruitment of lung units or by reduced or redirected blood flow within the injured lung. In the latter circumstance, PaCO₂ may also rise. When recruitment is the explanation for O₂ improvement, however, CO₂ exchange is not compromised and may even improve, reflecting increased alveolar ventilation. Similar principles apply during prone positioning.

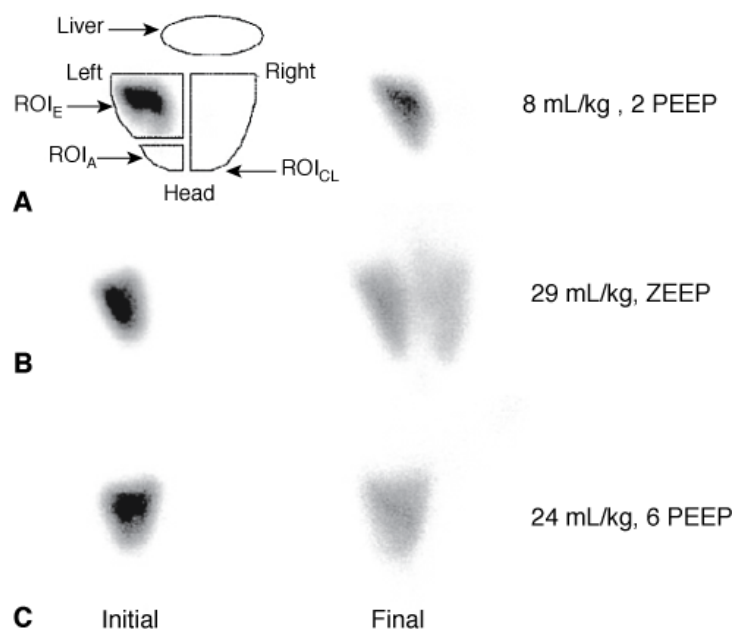
Table 29-7: How Should Positive End-Expiratory Pressure Be Adjusted? A Practical Compromise

Ensure adequate preload.
Use small to moderate tidal volume or driving pressure.
Recruit by increasing PEEP and/or pressure-controlled ventilation to plateau of approximately 50 to 60 cm H ₂ O for approximately 1 to 2 minutes.
Reduce driving pressure to 15 to 20 cm H ₂ O.
Reduce PEEP until arterial O ₂ or compliance falls significantly.
Re-recruit and select next higher PEEP if hemodynamics are acceptable.
Ensure that plateau pressure remains below acceptable maximum (e.g., <30 cm H ₂ O).

The prone position should be considered in patients with severe gas-exchange impairment regardless of their “recruiting test” result in the supine position. Patients requiring more than 10 to 15 cm H₂O PEEP at FiO₂ equal to or greater than 0.7 to maintain oxygen saturation at 90% or greater should be considered for prone positioning unless there is a clear contraindication or the patient is improving rapidly. The prone position should be considered independently of supine recruiting potential because as prone positioning will help lymphatic drainage and secretion removal and release the lower lobes of the lungs from the need to support the weight of the heart. Although provocative experimental data have challenged the concept,¹⁷² the preferred angle for head elevation in supine patients is 30 degrees to horizontal (Fowler) with frequent (at least every 2 to 4 hours) lateral turning. Similar rules apply in the prone position; reverse Trendelenburg at 15 to 30 degrees is preferred to flat (0 degrees) horizontal. Tidal volume is adjusted to the same value used in the supine position. An increase of plateau pressure strongly suggests that chest wall compliance has been altered by proning. In those instances, a proportional increase of PEEP may also be justified. (For example, if plateau pressure rises by 10%, PEEP would be increased by 10%.)

Prevention of Injury Propagation

Patients in the first stage of localized lung inflammation, such as lobar pneumonia, may have mobile secretions and edema that are both initially contained regionally rich in inflammatory mediators and surfactant-inhibiting protein. These injurious biofluids temporarily have the potential to spread via the airways to other healthy regions, propagating tissue damage (Fig. 29-23). Measures to contain them include use of moderately high PEEP, small tidal volumes and modest gravitational inclination, as well as the avoidance of forceful expiratory efforts. Although currently unproven, it seems prudent to take such “lung protective” measures during the earliest phase of ventilation so as to avoid the blossoming of focal inflammation into ARDS.¹⁷³



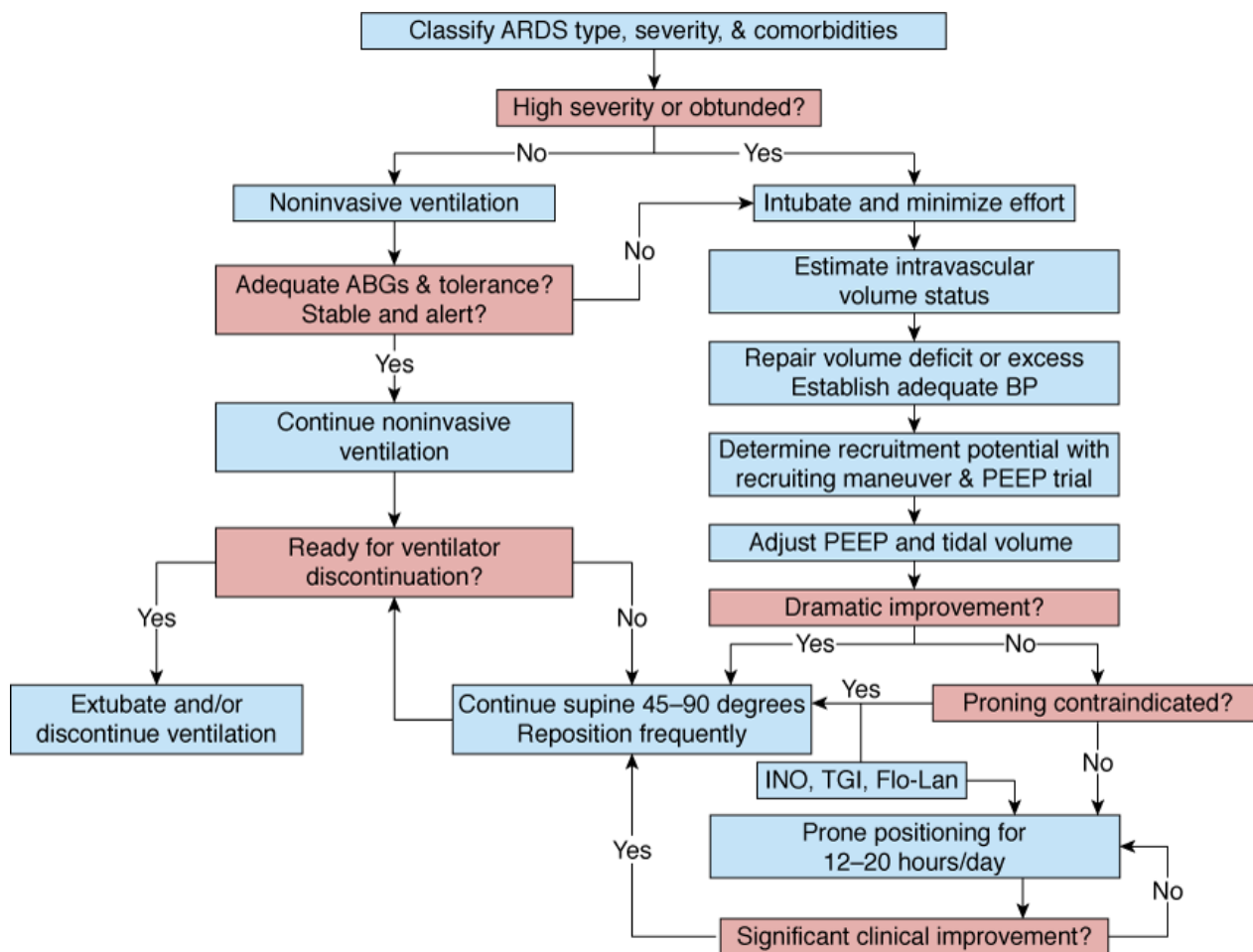
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Propagation and limitation of intraalveolar biofluid via the airway network. In this experimental small-animal model, radiolabeled albumin can be spread from its initial lobar position throughout the lungs by high tidal volumes and low levels of PEEP. Conversely, moderate PEEP contains the fluid to the lung in which it was instilled. *ZEEP*, zero end-expiratory pressure. (Adapted, with permission, from de Prost N, Roux D, Dreyfuss D, Ricard JD, Le Guludec D, Saumon G Alveolar edema dispersion and alveolar protein permeability during high volume ventilation: effect of positive end-expiratory pressure. *Intensive Care Med*. 2007;33(4):711–717.)

Suggested Sequence of Management Decisions

Figure 29-24 is a suggested algorithm for decision making in ARDS ventilation.

Figure 29-24



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A suggested decision sequence for ventilation decisions in ARDS. *ABGs*, arterial blood gases; *BP*, blood pressure; *Flo-Lan*, proprietary name for prostacyclin; *INO*, inhaled nitric oxide; *PEEP*, positive end-expiratory pressure; *TGI*, tracheal gas insufflation. (Used, with permission, from Marini and Gattinoni.⁵)

Initial Phase of Stabilization and Support

1. Determine whether the patient with oxygenation impairment has ALI, and if so, determine whether it is complicated by such reversible comorbidities as volume overload, pleural effusion, abdominal distension, or pneumothorax.
2. Initiate ventilation with face mask or intubate as severity warrants.
3. Decide on controlled versus spontaneous ventilation. Use controlled or nearly controlled ventilation to subdue respiratory efforts for the most severely involved patients during the early stage of support.¹⁷⁴
4. Initial ventilator settings: FiO_2 0.8, PEEP 5 to 8 cm H_2O (depending on concern regarding hemodynamic tolerance); tidal volume 6 to 10 mL/kg predicted body weight (depending on inspiratory plateau pressure and minute ventilation requirement).
5. Estimate volemic status initially from arterial blood pressure, respiratory variations of pulmonary and systemic arterial pulse pressure, central venous pressure, urinary output, and urinary electrolytes.
6. Confirm adequacy of intravascular volume using echocardiography, results from a volume challenge, and central venous and pulmonary artery catheter data (cardiac index, mixed venous O_2 saturation, and occlusion pressure), if available.
7. Replete any volume deficits and support the circulation with vasopressors and inotropes to the extent necessary to perform the ventilatory manipulations safely.

8. Determine the recruitment potential of the patient by using a recruiting maneuver and/or decremental PEEP trial.

9. Selected tidal volume should be inversely proportional to individual thoracic compliance and may range from 5 to 9 mL/kg of ideal body weight. During the PEEP trial, consider together the oxygenation change, the PaCO_2 change, the alterations of mechanics, and the hemodynamic response. Adjust the PEEP and tidal volume combination to the lowest tolerated values that sustain the recruitment benefit.
10. Proning is advisable in those with no contraindication and severe disease regardless of recruiting test, unless they are already improving rapidly. If the patient does not respond to the prone position, another recruiting maneuver is attempted while the patient is prone. The PEEP and tidal volume combination is readjusted as before.
11. When proning is used, scheduled reversion to the supine position is conducted at least once a day for cleanup, dressing changes, edema clearance, diagnostic procedures, transport to imaging, and so on, unless supine position is not physiologically tolerated. Many patients require almost continuous proning to maintain adequate gas exchange during the first several days of illness. Proning can be discontinued when it no longer makes an impressive contribution to oxygenation and plateau pressure can be kept in a “safe” range when the patient is supine.

Subsequent Care

Recovery onset is recognized by improving PaO_2/FI_2 and minute ventilation-to- PaCO_2 ratios, clearing radiographic opacity, and rising compliance of the total respiratory system. Appropriate adjustments are then made to sedation and ventilating pressures and spontaneous breathing encouraged by conversion to pressure-support ventilation or to PCV with lower driving pressures as tolerated. Proning is discontinued when the alterations in PaO_2 observed during position changes are less than 10%, status is clearly improved, or no obvious benefit has been achieved after a lengthy trial (>48 hours). Reductions in FI_2 to an acceptable range are undertaken before cutbacks of PEEP.

PEEP weaning is initiated when FI_2 is 40% or less and PaO_2 is 80 mm Hg or greater.

Future Directions and Research

Numerous questions related to the practical application of mechanical ventilation remain incompletely answered. These include the following.

Pathogenesis and Detection of Ventilator-Induced Lung Injury

Experimental data suggest that numerous intersecting variables can influence the severity of VILI. For a given combination of PEEP and tidal volume, vascular pressure, breathing frequency, and body temperature appear to be important contributing variables in the expression of VILI. Although both mechanical and inflammatory disruption of the air–blood interface are likely to be in some part responsible, how mechanical forces causing structural alterations and signaling inflammation are not well elucidated. Markers for damaging patterns of ventilation are badly needed.

Long-Term Damage to Airways and Parenchyma

The determinants of alveolar and small-airway damage occurring in patients ventilated for ARDS are not clear—peak pressure, end-expiratory pressure, or both. Future outcome studies should target the incidence of obstructive, restrictive, and disordered gas exchange following recovery from ARDS.

Relationship of Ventilator-Induced Lung Injury to Multisystem Organ Dysfunction

Although laboratory studies have shown the release into the systemic circulation of gas, bacteria, and inflammatory mediators from injured lungs ventilated at high pressures with insufficient PEEP, the causative role of VILI in multiorgan failure must be investigated further for the clinical setting.

Ventilator-Associated Pneumonia

The incidence of ventilator-associated pneumonia is high in intubated patients, especially among those with ALI. Whether and how ventilator-related injury predisposes to ventilator-associated pneumonia needs to be clarified, and the measures necessary to reduce its incidence identified.

Environmental Modifications

Multiple factors have been shown in the laboratory to be important in the expression of VILI, including elevations of precapillary pressure, reduced left-atrial pressure, and elevated body temperature. Whether such factors play a role in the clinical setting is not clear. Do such cofactors as edema, FI_2 , vasoactive and antiinflammatory drugs, and plasma protein concentrations determine the severity of expression of VILI? Are there additive or synergistic interactions among these variables?

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Appropriate Patterns for Positive End-Expiratory Pressure, Tidal Volume, and Inspiratory Flow

How to select the most lung-protective combination of PEEP and tidal volume remains an area of active controversy. Although considerable progress has been made, the debate needs to be settled regarding the relative importance of limiting peak inflation pressure, optimizing PEEP, reducing driving pressure, or limiting tidal volume. Should arterial oxygenation, thoracic mechanics, or bedside imaging (as by electrical impedance tomography or acoustic mapping) be used to guide these selections? What are the added values of using strain ratio (tidal volume-to-functional residual capacity) and transpulmonary pressure calculations using the esophageal balloon to adjust ventilator parameters? The advisability of reducing inspiratory flow rate and minute ventilation to minimize VILI deserves to be studied carefully.

Recruitment Maneuvers

Sustained inflation with high airway pressure is an effective means of opening collapsed tissue and may be integral to selecting PEEP. The value of repeated recruiting maneuvers and the best means of doing so, however, have not been studied carefully. The effectiveness of recruitment maneuvers may well vary with the type or stage of ALI. Moreover, the incidence of lung rupture, hemodynamic compromise, and other complications requires documentation. How should such maneuvers be performed? In which patients? How often, and at what FiO_2 ? Whether sighs help maintain recruitment in patients with low tidal volumes and unstable airways should be clarified.

Prone Positioning

The duration of proning (how long each day, when to cease proning in each patient) remains unsettled. Most investigators favor proning for most of each day, at least until there are clear improvements in mechanics and gas exchange, allowing adequate arterial oxygenation with FiO_2 less than 0.50 to 0.60 at tolerated levels of PEEP in the supine position. Whether proning confers an advantage on key outcomes other than oxygenation, at what stage of illness, and in which categories of patient, are other questions that merit further investigation. The advisability and appropriate intervals for body repositioning need further definition.

Noninvasive Ventilation

Noninvasive ventilation has been successfully applied to problems of acute respiratory failure, including some patients with ALI and ARDS.^{95–97} Although vastly better than before, interfaces must be improved to ensure reliability and patient comfort. More information is needed to guide clinicians regarding optimal selection of patients, timing, equipment, and monitoring.

Therapeutic Value of Hypercapnia and Tolerance of Hypoxemia

Laboratory data suggest that hypercapnic intracellular acidosis protects against some forms of oxidant injury.^{134–136} Its therapeutic value for the clinical setting requires further investigation. Adaptation to hypoxemia occurs gradually in healthy individuals exposed to altitude and in many patients with chronically impaired oxygen exchange. Can patients who are acutely ill adapt to gradual imposition of hypoxemia? If so, how fast do they do so?

Spontaneous versus Controlled Ventilation

Strong arguments can be made for reducing ventilatory requirements and controlling ventilation in the first stages of ARDS management. The publication of a trial on use of paralytic agents¹⁰¹ has energized debate as to the advisability of ceding control of the ventilatory pattern to the severely affected patient. The circumstances under which this is advisable or contraindicated as well as the most desirable ventilator mode need careful study.

Newer Modes of Ventilation

The physiologic rationale for high-frequency ventilation in ARDS is strong. It is not yet clear, however, that it confers benefits in beyond those achieved with conventional ventilator-based lung protective strategies. Definitive comparisons of high-frequency ventilation versus lung-protective strategies using a conventional ventilator are needed to identify possible benefits and hazards, as well as to learn the best approach to using this technology in the clinical setting. Certain less commonly introduced modes for ventilating ARDS, such as neurally adjusted ventilator assist,¹¹⁰ proportional assist ventilation,¹¹¹ “noisy pressure support,”⁸⁶ APRV,¹⁰⁹ and high-frequency percussive ventilation,¹¹⁸ are of strong interest, but their eventual place in ARDS ventilator management has not yet been settled.

Value of Adjunctive and Pharmacologic Measures

Tested as isolated interventions, [nitric oxide](#), partial liquid ventilation, aerosolized surfactant, corticosteroids, almitrine, and prone positioning have not been demonstrated in multicenter trials to influence mortality except in highly selected patient cohorts.¹⁵⁶ The physiologic rationale for these measures, however, is strong for certain types of patients and for certain clinical conditions. Moreover, few of these adjuncts have been tested in combination. (Prone positioning, for example, adds to the oxygenation benefit of inhaled [nitric oxide](#), but the impact of oxygenation of inhaled [nitric oxide](#) on survival has not been addressed.) The utility of such adjunctive measures—used alone and in combination—requires further study in populations most likely to benefit.

Summary and Conclusion

Injury to the alveolar–capillary membrane originates from diverse causes that may initially disrupt either the epithelial or endothelial surfaces. Managing the spectrum of conditions grouped together under the all-encompassing label of ARDS is a challenging assignment that must take into account the severity and stage of illness, as well as the interactions among the ventilatory and nonventilatory variables that influence the progress of this condition. Prevention of excessive tissue strain during the tidal cycle entails the avoidance of intolerable peak transalveolar and driving pressures, repetitive opening and closing of unstable lung units, as well as attention to pathogenetic cofactors that may initiate or exacerbate lung injury or retard healing. Because of the diversity of mechanical problems presented by different patients as well as the heterogeneity of mechanical properties among the lung units within the same individual, a logical approach requires mastery of key physiologic principles, awareness of the guidelines provided by well-conducted clinical trials, and integrated clinical judgment at the patient's bedside.

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